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14. ABSTRACT

This document provides the standard for designing and conducting tests estimating penetration of chemical agent vapor and aerosol simulant through chemical/biological (CB) protective suit systems while worn. Methyl salicylate (MeS) is used to simulate chemical agent vapor challenge. Nontoxic, fluorescent-tagged silica aerosol particles are used to simulate aerosol chemical agent challenge. For the vapor tests, personal sampling devices (PSDs) such as passive absorbent devices are used to monitor chemical concentration inside the suits. For the aerosol tests, a discussion of sampling methods to determine chemical concentration inside the suits is given. Test data are analyzed using a body region analysis procedure (BRAP) to provide an indication of the protection levels of the suit systems when exposed to vesicant or nerve chemical agents.

15. SUBJECT TERMS
Aerosol, Man-In-Simulant Test, MIST, methyl salicylate, MeS, passive absorbent device, PAD, Body Region Analysis Process, BRAP, whole suit system testing, Chemical/Biological, CB, Defense Testing.

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US ARMY DEVELOPMENTAL TEST COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure (TOP) 10-2-022 **14 December 2005**
DTIC AD No. : ADA440290

CHEMICAL VAPOR AND AEROSOL SYSTEM-LEVEL TESTING OF CHEMICAL/BIOLOGICAL PROTECTIVE SUITS

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1. SCOPE

a. This document provides the standard process for designing and conducting tests estimating penetration of chemical agent vapor and aerosol simulant through chemical/biological (CB) protective suit systems while worn. Suits may have any combination of the following characteristics:

(1) Suits may be constructed of air permeable, semipermeable, or impermeable fabrics.

(2) Suits may be of single or multilayered design.

(3) Suits may be constructed of inert or sorptive fabrics.

b. This TOP describes standard procedures for testing complete suit systems of protective clothing worn by test participants (TPs) when exposed to vaporized and/or aerosol chemical warfare agent (CWA) simulants. These procedures test the entire ensemble, including seams; closures; and interfaces at the ankle, wrist, waist, and neck.

c. The test procedures described in this document must be referenced and/or incorporated into a detailed test plan (DTP) or similar document. These procedures may be modified in the DTP to accommodate unique items or materials, or to satisfy specific testing requirements specified in the system evaluation plan (SEP) or other acquisition document. Alteration, however, will be made only after full consideration of how the changes may affect the reliability and validity of the data. These alterations, a description of the effect desired by the change, and the changes in the assessment process must be fully described in the DTP.

d. The consideration process will include a risk assessment addressing the impact to the following areas associated with each change and will be coordinated in advance with the organizations concerned.

(1) Safety.

(2) Test conditions.

(3) Environmental.

(4) Human use.

(5) Data quality.

(6) Test validity.

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1.1 Objectives

a. This TOP provides information necessary to plan, conduct, and report CWA vapor and aerosol testing of CB protective suits and discusses required facilities, equipment, procedures, test and experimental parameters, and data obtained using these test methods.

b. This document describes the testing methods currently in use for CWA vapor and aerosol testing of CB protective suits and the specific or range of test parameters required for each method. Usually, several tests and particular test parameters must be chosen from among those listed herein. The test parameters chosen for a particular test program may change depending upon objectives.

c. This TOP is to be used as a guide in preparing program-specific test plans or DTPs.

(1) Procedures described in this document may require tailoring to address the particular purpose and requirements for a specific protective suit.

(2) Any modification(s) of these procedures and the specific parameters used must be described in the DTP with the rationale for the modification(s) and an analysis of the possible ramifications of such alterations.

(3) The DTP should describe the specific test methods and parameters to be used. These may be based on factors such as the concept of operations requirements and/or threats to the protective suit being tested.

d. This TOP provides the procedures for characterizing protective suit performance as a baseline for quality assurance (QA).

1.2 Limitations

a. Data obtained by these procedures cannot be correlated to specific field conditions.

b. The DTP may require modification for unique test items or materials to satisfy specific testing requirements as may be specified in a SEP. However, alteration of the procedures contained herein shall be made only after full consideration of the possible effects the changes may have upon reliability and validity of the data. Alterations to this TOP will be coordinated in advance among all concerned organizations as part of test planning.

c. This TOP provides guidance regarding test design issues and data requirements that should be augmented by information found in the SEP and test evaluation master plan (TEMP) and described in the DTP. For those testing programs in which a SEP is not available or not applicable, the test director (TD) should consult with the customer and use previous documents as a guide in addition to this TOP.

d. This TOP is limited to approved standards and procedures. Developments in practices, equipment, and analysis may necessitate establishing new testing baselines. Additionally, standards of performance must be adjusted with the development of new technologies. Test procedures and parameters listed in this TOP require updating to accom-

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modate new technologies in test items or in test instrumentation. Any updates should be described in the DTP.

2. FACILITIES AND INSTRUMENTATION

2.1. Facilities

2.1.1 Man-in- Simulant Test (MIST)

Item	Requirement
Medical Treatment Facility.	**1. A medical facility with individuals trained and equipped to treat overexposure to CWA simulant or adverse reactions to physiological stress must be available.** **2. A method for measuring heart stress must be available.** **3. Trained individuals will include emergency medical technicians (EMTs) qualified in advanced life support.** **4. EMTs will be present during all CWA simulant trials.** **5. The EMTs will watch for possible adverse physiological responses in TPs and provide appropriate medical aid whenever necessary.**

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Test Chamber.	**1. The test chamber must be able to prevent simulant vapor contact with the environment or areas of the facility not intended as exposure areas.** **2. The test chamber must be large enough to hold at least two TPs and one on-floor supervisor (TP monitor) simultaneously.** **3. The chamber must have sufficient room for any exercise equipment and prescribed physical activity.** **4. The chamber must contain fans capable of providing a stable, uniform airflow directed toward the TPs at a variety of wind speeds from 3.2 to 16.1 km/hr (2 to 10 mph).** **5. Chamber temperature must be controlled at 21.1 to 32.2°C (70 to 90°F) and RH at 50 to 90 percent.**
Operator's Area.	**1. The operator's area will be occupied by test conduct personnel who will adjust wind speed, temperature, and RH and operate the simulant vapor generator and the chemical vapor concentration measuring instruments.** **2. This room will also have some means of visually observing the TPs and communicating with the on-floor supervisor.**
Chemical Laboratory.	**The chemical laboratory will provide the general analytical laboratory support needed for work with CWA simulants, including sampler analysis, instrument standardization, and hazardous waste disposal.**
Test Safety and Control System.	**The core body temperature and heart rate of TPs will be monitored during the trials. The personal vital signs monitoring system (PVSMS) is a system suitable to this task, but other methods are acceptable if approved by appropriate medical and human use authorities.**

2.1.2 Aerosol

Item	Requirement
Test Chamber.	**1. The test chamber must be able to prevent CWA simulant aerosol contact with the environment or areas of the facility not intended as exposure areas. While aerosol is generated in the exposure chamber, the air pressure must be maintained below the air pressure is surrounding rooms to avoid contamination in the adjacent rooms.** **2. The exposure chamber must be large enough to hold the number of TPs per test challenge required by the SEP and the DTP.** **3. The chamber must contain a fan capable of providing a stable, uniform airflow directed toward the TPs at a variety of wind**

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	speeds from 0 to 18 m/s (0 to 40 mph).
	4. Chamber temperature must be controlled between 16 and 32°C (60 and 90°F) and RH between 20 and 80 percent within ±10 percent of target value.
Operator's Area.	**1. The operator's area will be occupied by test conduct personnel who will adjust wind speed, temperatures, and RH and operate the CWA simulant aerosol generator and the chemical aerosol concentration measuring instruments.**
	2. This room will also have some means of visually observing and communicating with the TPs (communication may be nonverbal, such as via computer monitor and hand signals).
Chemical Laboratory	**The chemical laboratory will provide the general analytical laboratory support needed for work, with CWA simulants, including sample analysis, instrument standardization, and hazardous waste disposal.**

2.2 Equipment

2.2.1 MIST

Item	Requirement
Vapor Generator.	**1. The generator must be capable of operation by remote control from the operator's area and must be able to dispense MeS at the controlled rate required to maintain vapor concentration at any value between 10 and 1000 mg/m^3 ±5 percent).**
	2. The system must provide near real-time data collection and control to allow timely adjustments of control devices within their operational limits.
	3. An automated data collection system (Paragon® Automatic Data Collection System software or equivalent) must be used to collect all real-time data. The automated control system software included in the Paragon® system, or equivalent, must be used to control the test chamber system.

2.2.2 Aerosol

Item	Requirement

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Aerosol Generator.	**1. The generator must be capable of maintaining aerosol mass concentration of 100 to 150 mg/m^3 and a particle size distribution between 0.1 and 10 µm within the test chamber for up to 1 hour of continuous operation. The aerodynamic mass median diameter must be 2 to 3 µm and the geometric standard deviation must be between 2 and 3.** **2. The system must provide near real-time control to allow timely adjustments of control devices within their operational limits.**

2.3 Instrumentation

2.3.1 MIST

Devices for Measuring	Measurement Accuracy
Test chamber atmospheric temperature.	**±0.5°C at 38.5°C. The device must be able to sample at least once every 2 minutes.**
Test chamber RH.	**±2.5 percent**
Test chamber wind speed.	**±2.5 percent**
Real-time monitors (RTMs), or near RTMs, for monitoring atmosphere challenge concentration of CWA simulant vapor. A Miniature Infrared Analyzer® (MIRAN®) has been used for this function and meets the requirements.	**±2.5 percent over the range of 10 to 1000 mg/m3 for each instrument.**
Uniformity of vapor chamber chemical concentration.	**±15 percent averaged over all instruments across the chamber.**
In TP's dress and undress areas, an RTM more sensitive than the MIRAN®, such as the Miniature Automatic Continuous Air Monitoring System® (MINICAMS®), may be used.	**±5 percent of the lower detection limit (LDL)of the MIRAN® for MeS, approximately 0.05mg/m3.**
A PVSMS or equivalent system used to monitor the TP's core temperature and heart rate.	**±1.0 percent.**

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2.3.2 Aerosol

Devices for Measuring	Measurement Accuracy
Test chamber atmospheric temperature	±0.5°C accuracy in the test range of temperatures. The device must be able to sample at least once every 5 minutes.
Test chamber RH.	±10 percent.
Test chamber wind speed	±0.45 m/s (±1.0 mph).
Challenge aerosol mass concentration.	At least 98 percent efficient, by mass, at collecting the challenge aerosol.
Aerodynamic particle size distribution. A multistage cascade impactor has been used in previous aerosol challenge tests.	The desired range of aerosol particle size is between 0.5 and 10 μm.

3. REQUIRED TEST CONDITIONS

3.1 MIST

3.1.1 Test Planning

a. Experimental Design

(1) The test will be designed to facilitate data analysis using standard techniques.

(2) The candidate ensembles will be randomly distributed among the trials to minimize the effect of any trial-to-trial variation in the overall analysis. Individual trial matrices will be provided under separate cover.

(3) The chamber position and rotation order of each TP must be specified. Rotation into and from the test chamber will be conducted in a manner that prevents introducing bias for any particular suit based on chamber position or rotation order.

(4) The exercise stations and exercise activities are in Appendix C.

b. The TD will have all pertinent documentation available. Documentation includes:

(1) Safety release and approval from the authorizing agency [U.S. Army Developmental Test Command (DTC), Aberdeen Proving Ground, Maryland] to begin testing.

(2) Human Use Committee (HUC) approval or exemption and notification.

(3) Government and manufacturer's publications, including the current material safety data sheet (MSDS) for MeS.

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(4) Requirements document.

(5) SEP (if applicable).

(6) Safety assessment report (SAR).

(7) Test planning or execution directive.

(8) System support package (SSP) and SSP list (SSPL).

(9) Environmental impact assessment for life cycle (EIALC).

(10) National Environmental Policy Act (NEPA) documentation for the test. This may be a record of environmental consideration (REC), environmental assessment (EA), environmental impact statement (EIS), or other NEPA documentation as required.

(11) Other documentation as necessary [e.g., TOPs, SOPs, etc.].

c. Familiarization

(1) Potential problem areas must be identified by reviewing previous records and results of similar tests.

(2) Relevant SOPs and other procedures will be reviewed for applicability, completeness, and adequacy. These documents will be updated as required.

(3) Development of DTPs will require:

(a) Review of the applicable SEP and other test guidance literature.

(b) Familiarization with preceding development and test phases.

(c) Consideration of data from previously conducted tests in order to avoid duplication and to reduce the scope of further testing.

d. EA

(1) In compliance with NEPA, the Department of the Army requires that an EIALC be prepared and that ptential environmental impacts be assessed at the earliest possible stage in the planning process of any new equipment.

(2) Testing at DTC facilities must also be assessed for environmental impact.

(3) A detailed EIS will be prepared by the test center and evaluated in accordance with (IAW) NEPA processes when the proposed action may significantly affect the environment, is environmentally controversial, or when litigation is expected based on environmental issues.

(4) A REC will be completed for the test if review indicates that there is existing NEPA documentation in place for the action or there is an applicable categorical exclusion.

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The REC will indicate the process for consideration of the test and rationale for the conclusion.

(5) The TD will ensure that a life cycle EA, an EIS, or other appropriate documentation has been received and understood before the test begins.

3.1.2 Test Equipment

a. Instruments, samplers, and equipment used for this test must meet strict criteria (Appendix A, Paragraphs 1.1, 1.2, and 1.3). All instrumentation will be required to undergo a validation process.

b. Passive absorbent devices (PADs) are small sampling devices designed to evaluate the total dose of simulant received at a specific location. The device is a small packet filled with Tenax® absorbent material backed with adhesive to enable the device to be placed directly on the skin. Ten percent of the total number of PADs used in the test must be available for use as controls.

(1) The PAD sampler, developed by U.S. Army Natick Soldier Research, Development, and Engineering Center (NRDEC), Natick, Massachusetts, was selected as the best passive absorbent sampler for the MIST. The selection criteria included the following:

(a) Sensitivity.

(b) An agent uptake rate similar to skin.

(c) Size and shape that did not interfere with normal movements and suit configuration.

(d) Easy adherence to human skin.

(2) PADs are small samplers that use diffusion to collect chemical vapor from the air. These samplers do not pump air, which is an advantage because no tubing or wiring breaches the suit system and no air circulation is induced by the sampling process.

(3) PAD Packet

(a) The PAD sampler is a small packet [approximately 3.8 by 3.2 by 0.3 cm (1.5 by 1.25 by 1/8 in)].

(b) The exposed face of the packet is a 0.025-mm (0.001-in) thick high density polyethylene film that acts as the diffusion barrier.

(c) The back of the packet is an aluminum foil/polyethylene/nylon membrane impermeable laminate.

(d) Identical quantities of Tenax® TA absorbent are sealed inside each packet.

(e) The samplers are sealed inside foil/plastic laminate (impermeable) packages by the manufacturer to prevent exposing the sampler before use.

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(f) A coating of medical adhesive on the back of the packet allows attachment directly to the skin of the TP.

(4) The mass of chemical absorbed is proportional to:

(a) The vapor concentration.

(b) The rate of chemical diffusion through the diffusion barrier, also known as the uptake rate.

(c) The area of the sampling surface.

(d) The exposure time.

(5) PADs usually require a significant concentration of chemical vapor in order to register a detection; therefore, they are suited for use in cases where large concentrations of the chemical are expected.

(6) PADs estimate the total amount of chemical absorbed over the entire sampling period. This estimate can be converted into the average concentration of agent present in the vapor during the sampling period.

(7) Upon receipt, if all samplers in the population have identical histories (have been taken from the same source, have been stored together, have been treated identically, etc.), these samplers must be subjected to a lot acceptance test designed to determine if the samplers are suitable for use.

(8) The sampler must collect chemical vapor at a specified rate, stick to TPs, and be free from contamination in order for the sampler to operate properly.

(9) The following tolerances are proposed for lot acceptance:

(a) Sampler area: 2 percent RSD.

(b) Weight of absorbent: 10 g to ±25 percent.

(c) Sampler uptake rate (at one concentration): 5 percent RSD.

(d) Spike recovery (at one concentration): 95 percent.

(e) Chemical cleanliness: 3 ng interferents.

(10) Samplers must be handled in such a way as to avoid contamination. Handlers must wear new latex gloves when handling the samplers, avoid touching the sampling surface, work in a clean environment (preferably a clean box), and use tweezers whenever possible.

(11) Upon removal from the TP, the samplers must be attached to individual pieces of aluminum foil with dimensions greater than that of the sampler.

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(12) Foil-backed samplers must be placed in individual airtight vials. The vials must be placed in a refrigerator at 4 to 6°C (40 to 42.8°F) within 1 hour.

(13) The samplers must be analyzed (Appendix A, Section 3.2.f) within 24 hours of collection or stored in a freezer for no longer than 30 days.

c. Chamber Concentration Monitors

(1) RTMs, such as MIRAN®s, are required for monitoring MeS concentration in the test chamber atmosphere.

(a) The MIRAN® is an RTM that measures absorption of infrared energy using an infrared spectrometer. The instrument includes a gas cell with a volume of 5.6 L and a variable path length ranging from 0.75 m (2.5 ft) to 20.25 m (66.4 ft). The spectrometer provides a chemical vapor concentration estimate as air is drawn through the gas cell.

(b) When vapor concentration changes, the response time of the MIRAN® lags behind until the vapor concentration in the sampling cell reaches the new concentration. Lag time can be controlled by adjusting the vapor flow through the instrument.

(c) The instrument has a relatively high LDL and is suitable primarily for monitoring the challenge concentration. The LDLs of the MIRAN® 1A for four selected simulants are given in Table A.1.

(2) The number and location of RTMs required must be determined by test chamber mapping performed during the pretest mapping phase (Appendix A, Paragraph 6.2).

(3) If air from the chamber will be drawn into the RTMs by a pump, the tubing must have a low absorption factor for the simulant. Fluorinated polymers are resistant to chemical absorption. Examples of acceptable flexible tubing are: Teflon® fluorinated ethylene propylene (FEP); Teflon® perfluoroalkoxy (PFA); Teflon® polytetrafluoroethylene (PTFE) (all from Cole-Parmer, Vernon Hills, Illinois); and Pureline II® (Dixon Industries, Charlotte, North Carolina).

(4) The pressure inside each RTM while drawing air must not differ from the ambient air pressure by more than 2 percent.

(5) The RTMs will be calibrated for MeS concentrations over a range such that the desired challenge concentration is mid-range (i.e., a range of 0 to 200 mg/m^3 for a 100-mg/m^3 challenge).

Table 1. Typical Lower Detection Level (LDL) for Miniature Infrared Analyzer® 1A for Selected Chemical Simulants for a 20.25-m Path Length and Defining the LDL at 0.005 Absorbance Units; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

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Chemical	Analytical Wavelength (m)		LDL (mg/m^3)
Diethyl malonate (DEM)	8.8	(28.9 ft)	1.9
Methyl salicylate (MeS)	8.3	(27.2 ft)	1.7
Dimethyl methylphosphonate (DMMP)	9.5	(31.2 ft)	0.9
Sulfur hexafluoride (SF_6)	10.7	(35.1 ft)	0.2

d. The clear areas must be monitored. An RTM, such as a MINICAMS®, will be required for monitoring the dress and undress areas for MeS concentration. The number and location of monitors required must be specified for each test.

(1) The MINICAMS® is a near RTM containing a solid sorbent trap; a gas chromatograph (GC); and the connections, ovens, and other equipment necessary to sample an airstream for a selected interval of time.

(2) Analyte is collected and concentrated using a solid sorbent tube. Once collected, the analyte is then thermally desorbed from the tube into the GC for chemical analysis.

(3) According to the manufacturer's operation manual, the MINICAMS® can measure CWA vapors in air to a level meeting the Surgeon General's 8-hour time-weighted average concentration when operating on the 3-minute cycle (1 minute of air sampling at 1 L/min). Longer cycle times provide longer air sampling periods and lower detection levels.

(4) The MINICAMS® is capable of giving near real-time estimates of chemical concentration present in the air sampled. One MINICAMS® can be dedicated to sampling a single area.

(5) In addition, an automated valving system using a single MINICAMS® can also be used to sequentially sample a series of test areas. Such sequential sampling increases the time between data points obtained from any single test area but provides near real-time information across several test areas.

(6) Suitable care must be given to the choice of sampling lines, airflow rates, and sequencing valve construction materials in order to minimize chemical loss through sorption onto surfaces and time delays caused by stagnant air in the sampling lines leading to the sequencing valve.

(7) The instrument can be rendered unresponsive to changes in chemical concentration due to saturation by exposure to high concentrations. In such a case, time must be allowed for the instrument to purge and recover.

e. A core temperature and heart rate monitoring device, such as a PVSMS, will be required for monitoring the safety of each TP.

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(1) The PVSMS is a small transistor-based system that relays physiological data about each TP to a computer-based data collection and readout station used in the MIST. This system will allow EMTs and test control personnel to monitor each TP (in the MeS environment) for heat stress and other physiological signs of distress.

(2) The system measures:

(a) Core body temperature by use of a small, pill-sized monitor (which is swallowed).

(b) Heart rate by use of a heart monitor strapped around the TP's chest.

(c) Skin temperature taken by a probe attached to the skin next to the heart monitor.

(d) Skin temperature taken by a probe attached to the skin on the TP's side next to the suit and the heart monitor harness.

(e) Skin temperature taken by a probe attached to the skin underneath each of the TP's arms.

(3) These readings are transmitted to the personal data collection instrument package that is attached to the TP's left side, outside the suit. The instrument package will be maintained on the outside of the TP's clothing in a manner that minimizes interference with the test, such as in a garment pocket on the TP's left arm or taped to the upper left arm.

(4) A radio frequency antenna that is mounted inside of the test chamber will transmit the data from the pill receivers to a data collection system and an auxiliary station inside the control room.

f. <u>Physical Environment Monitors</u>

(1) The temperature, RH, and wind speed inside the chamber must be monitored.

(2) The number and location of the temperature, RH, and wind speed sensors required to adequately monitor the chamber must be determined by test chamber mapping (Appendix A, Paragraph 6.2).

g. <u>Communication</u>

(1) If the protective suit does not include a communication system, or if the communication system within the suit is incompatible with available communication systems, a means must be provided for the TD and the TPs to communicate with each other.

(2) The on-floor supervisor must be in direct communication with the test control center at all times.

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3.1.3 Test Preparations

a. Several actions will be required before the start of testing to ensure reduction of test bias on the part of facilities or instrumentation. These actions include:

(1) Testing and certification of:

(a) The test chamber.

(b) The instruments and instrument locations.

(c) The sampler inlet locations in the test chamber.

(2) Verification of the sampling and analysis process.

(3) Background testing of the TPs, masks, and other test items.

b. Test chamber instrumentation will be calibrated IAW recommended manufacturer's and local procedures, local SOPs, and QA/QC procedures.

c. An ORI will be conducted in two steps (Appendix A, Paragraph 3.3.b) before the start of the actual test to ensure that all system components are performing adequately.

d. Requirements and criteria for the test chamber and test instrumentation are outlined in Appendix A, Paragraphs 1.1, 1.2, and 1.3.

e. PADs are placed on the bodies of the TPs at the locations shown in Figure A.1 and described in Table A.2.

f. Test Items

(1) The following steps will be taken during the receipt inspection:

(a) The interior and exterior of each item of clothing must be inspected for rips, tears, and other damage.

(b) Each suit component will be assigned and marked with an individual TIIN. Care must be used to ensure that the TIIN is readily visible, does not compromise the integrity of the test item, nor interfere with sampling methods.

(c) Each ensemble will be given a TICN IAW the overall TICN numbering system. The TICN assigned to each item will uniquely identify it by test program and be used to track the item throughout the test process.

(d) Photographic records (with metric scale and in focus) of items showing damage will be made with a written record of the damage. This record will show all areas of damage and will be cross-referenced to the recorded TIIN and TICN. Receipt inspection will be carried out as closely as possible to the time of the test.

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(e) If worn suits are to be tested, the wear history of each must be provided and recorded in the chemical test database.

(f) Required maintenance operations described in applicable technical manuals will be performed and all mechanical components tested for operability.

(2) Any pretest environmental conditioning of the clothing/suit required by the SEP and the DTP will be performed.

(3) After the receipt inspection, the suits will be stored in a manner preventing contamination by any chemical vapors present in the storage area.

(4) Suits must be stored so as to ensure degradation does not continue after wear. These conditions should be stated in the SEP and must be specified in the DTP. Special storage conditions must include a room temperature of 5°C and ambient RH.

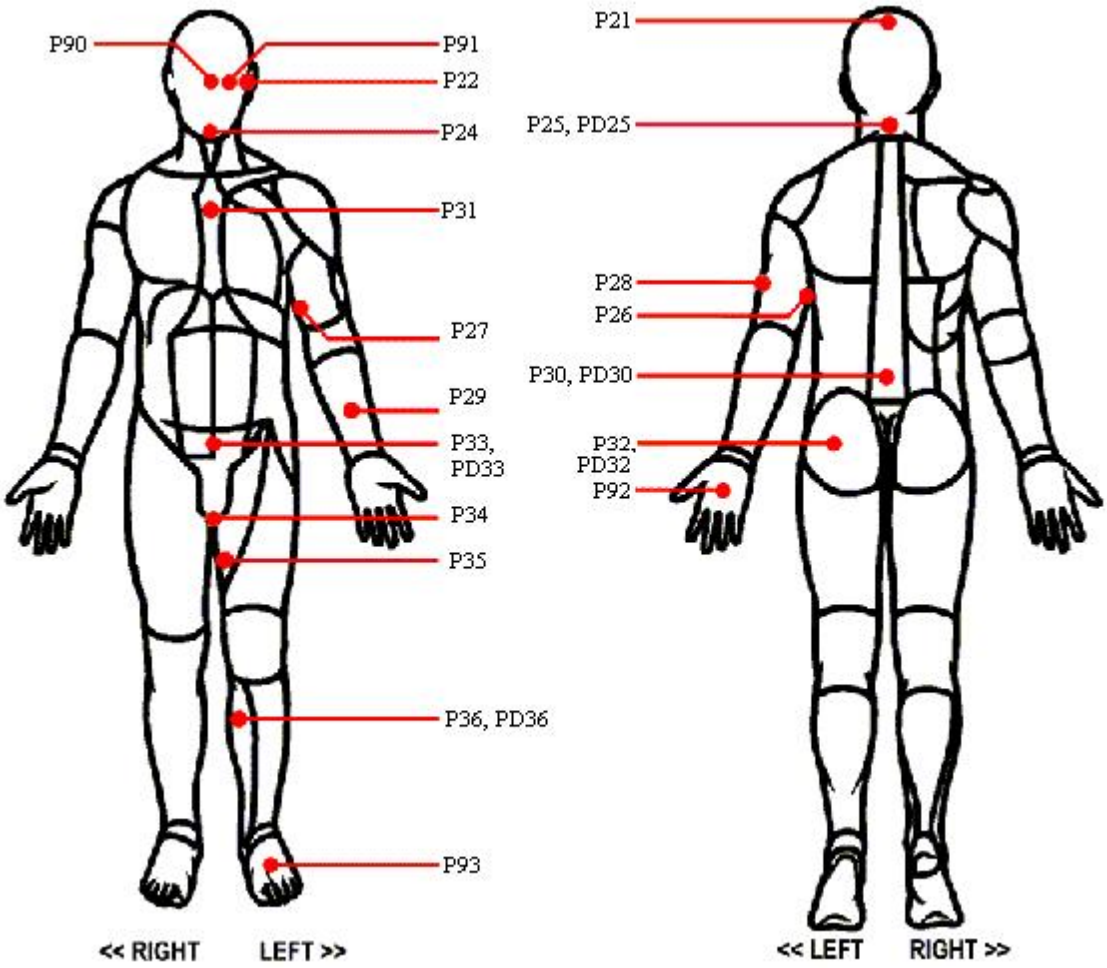


Figure 1. Sampling Locations for the Passive Absorbent Devices (PADs) for Man-In-Simulant Test (MIST); Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

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Table 2. Passive Absorptive Device (PAD) Placement Location Descriptions; Individual Protective Equipment for Selectively Permeable Membrane Materials Man-In-Simulant Test.

Position Number	Description
P21	Scalp
P22	Left Ear
P24	Chin
P25, PD25	Nape
P26	Armpit
P27	Inner Upper Arm
P28	Outer Upper Arm
P29	Forearm, Volar
P30	Mid Back
P31	Abdomen
P32, PD32	Buttocks
P33, PD33	Groin
P34	Crotch/Scrotum
P35	Inner Thigh
P36, PD36	Inner Shin
P37	Blank Transport Sample for Quality Control (QC)
P38	Spike Transport Sample for QC
Don	Dress Area
Doff	Undress Area
P90	Nose Cup
P91	Mask
P92	Glove (Hand)
P93	Boot (Foot)

See Figure A.1.
Indicates duplicate PSD at these locations.
PSD placed in transport case to monitor cross-contamination.
[D]PSD placed in mask, not on the test participant's body.
[E]PSD placed in appropriate dressing area.

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(5) Suits requiring cleaning or laundering before use will be washed in an approved soap which will not interfere with the sampling procedure.

(6) The TD will ensure that the suit developer provides new equipment training (NET) whenever necessary.

g. Contamination and interferent control begins with the TPs; therefore, TPs will be advised of the following:

(1) TPs will be provided with instructions about foods and products containing MeS that must not be consumed or used within 48 hours before testing begins or during testing.

(2) TPs will be required to shower with MeS-free soap and dress in MeS-free clothing.

(3) Background sampling will be conducted to screen for interferents and to train TPs in likely areas of contamination.

h. Dress and Undress Areas

(1) The dress and undress areas will be washed with MeS-free detergent before use to remove any MeS or other potential interferents. RTMs (MINICAMS®) and PADs (Table 3) will be used to sample for background levels of MeS or interferents. The dress and undress areas must be kept free of MeS or MeS vapor in the air.

(2) A passive sampling device (PSD) and/or low-level RTM (MINICAMS®) will be used to monitor the dressing and undressing areas for low levels (0.25 mg/m^3) of simulant or interferents that may be picked up during placement or recovery.

(3) All TPs will be sized and fitted with the garments to be tested.

3.1.4 Preliminary Actions

a. Pretest actions are designed to ensure that the test is conducted in a test chamber with well-defined airflow and controlled temperature and humidity.

b. Interferent Controls

(1) The key factor in the success of this protocol is control of MeS and any other interferents that may contribute to testing inaccuracy.

(2) Before the start of each test, the test items will undergo sampling for any interferents present in the ensemble. Testing may be performed using either of two methods:

(a) The preferred method is to test each item before each trial.

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(b) Alternatively, a selected number of items representative of each configuration type will be tested. If all of the test items in a population have identical histories (have been taken from the same source, have been stored together, have been treated identically, etc.) and those selected are found free of interferents, the remaining suits of that type will be assumed to be free from either an interferent or background levels of MeS.

Table 3. Summary of Control Samples for Man-in-Simulant Test (MIST); Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

Type of Control Sample	Number of Control Samples	Purpose of Control Sample
Blank PADs and MINICAMS®	Two PADs for each test participant	To check for contamination or interferents in dressing and undressing rooms.
Blank PADs on exterior of Tyvek® suit	One full set	To check the RTM measurement of the MeS concentration in the chamber.
Blank PADs and spiked PADs	Two of each type of PAD per set of suits	Used as interferent controls.
Spiked PAD storage controls	Several	To check sample degradation or interferent pickup during storage.
Spiked laboratory controls	10 percent of the total number of samples	A check to determine if the chemical analysis is in control.
Standard solutions	Three/MIRAN®/day	To check the operation of the MIRAN®s.
Spiked PADs and blank PADs	Two each per trial or test participant	To check for contamination/loss during transport and testing.
Blank PADs and MIRAN®s	Determined by test director	To check the chamber for contamination or interferents before testing begins.

Passive absorbent devices.
Miniature Automatic Continuous Air Monitoring System®.
Real-time monitor.
Methyl salicylate.
Minature Infrared Analyzer®.

(c) The alternate method may be conducted either as a separate pretest before the start of all trials or as individual tests just before each trial and can be discontinued after a sufficient number of suits have been sampled.

(d) The alternate method has two limitations: (1) The screening must be continued or restarted if the potential exists for an increased interferent or background simulant level such as occurs if the items have been reused, and (2) groupings of test items must be carefully selected to ensure that a critical difference is not ignored between the tested items and the control items.

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(3) Background sampling will be performed in the following manner:

(a) One complete suit (or more) of each type will be placed in a nonpermeable bag or other suitable container, and the ensemble will be held for at least 2 hours at 32°C (90°F).

(b) The air in the bag/container will be sampled using one PAD for a length of time equal to the duration of each trial. (CAUTION: *Sample the empty bag/container for volatile interfering chemicals before sampling the suits.*)

(c) The PAD will be analyzed for MeS to determine whether the suit or the bag/container emits chemical(s) that could give a false positive reading. If positive results are obtained, each component (suit, hood and mask, boots, and gloves) will be tested separately to determine the source of the interferent or background contamination. A determination can then be made to replace the item or subtract the interferent/background results from the sample results.

(d) Another analytical method may be chosen, such as RTM sampling, which discriminates between MeS and off-gassing chemical(s).

(e) Air supplied to the bag during sampling must be passed through a carbon filter before entering the bag.

(f) A number of spiked controls (PADs that have absorbed a known amount of simulant) should also be used to determine if chemicals that react with or destroy MeS are released from the garments.

c. Test Setup

(1) The chemical vapor generator; temperature, RH, and wind speed instruments; and RTMs will be placed at the same locations as when the pilot test/ORI (or mapping phase) was conducted.

(2) The control and signal cables will be neatly arranged to lead from the test chamber to the operator's area. The cables will be clearly marked to identify the specific instruments to which each is connected.

(3) Samplers will be procured before the test begins.

(4) The samplers will undergo a lot acceptance test upon receipt.

(5) The RTMs will be standardized with MeS by a validated procedure, installed, and checked at the test chamber for correct response before testing begins.

(6) The instruments for measuring the temperature, RH, and wind speed in the test chamber will be calibrated and installed.

(7) The control-system software will continuously control all chamber functions.

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(8) The automatic data collection system software will continuously record the output of the RTMs, air temperature, air pressure, and RH measuring instruments.

(9) The control system and automatic data-collection system software will be connected to all equipment and test instruments and checked for readiness.

(10) The software used for data collection of nonreal-time data will be checked for readiness by performing the entire data-collection process using a test (dummy) data set.

3.2 Aerosol

3.2.1 Test Planning

a. Experimental Design

(1) The test will be designed to facilitate data analysis using standard techniques.

(2) Candidate ensembles will be tested in random order whenever possible. Some grouping of tests may be needed, depending upon availability of test garments and size-match to available TPs. Also, if testing garments which vary widely in their level of aerosol protection, some order in the test sequence may be necessary to prevent carry-over contamination from TPs wearing suits with high levels of aerosol penetration to subsequent tests with suits having low aerosol penetration.

(3) The exercise activities are given in Appendix B.

b. The TD will have all pertinent documentation available. Documentation includes:

(1) Approval from the authorizing agency (e.g., DTC) to begin testing.

(2) HUC approval or exemption and notification.

(3) Government and manufacturer's publications, including the MSDS(s) for the aerosol(s).

(4) Requirements document.

(5) SEP.

(6) SAR.

(7) DTP or execution directive of scope of work.

(8) SSP and SSPL.

(9) EIALC, if applicable.

(10) Test-specific NEPA documentation.

(11) Other documentation as necessary [e.g., TOPs, SOPs, etc.].

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c. Familiarization

(1) Potential problem areas must be identified by reviewing previous records and results of similar tests.

(2) Relevant SOPs and other procedures will be reviewed for applicability, completeness, and adequacy. These documents will be updated as required.

(3) Development of DTPs will require:

(a) Review of the applicable SEP and other test guidance literature.

(b) Familiarization with preceding development and test phases.

(c) Selection of appropriate methods, sequences, facilities, and test equipment.

(d) Consideration of data from previously-conducted tests in order to avoid duplication and to reduce the scope of further testing.

3.2.2 Test Equipment

a. Instruments, samplers, and equipment used for this test must meet strict criteria. All instrumentation must undergo a validation process.

b. Appropriate types and numbers of instruments will be selected for the size of the test chamber.

c. The following equipment, materials, and supplies are required and must be available at the test site.

(1) A fluorometer is required for the analysis of filters, impactor slides, and samples of aerosol rinsed from the TP's skin and garment. The fluorometer must be capable of measuring the fluorescence of a sample within ±10 percent. A Turner Fluorometer Model 112 or similar instrument is suggested. The fluorometer will be operated with excitation and emission filters appropriate for fluorometric analysis of uranine.

(2) Test Aerosol

(a) The test aerosol is a synthetic amorphous silica powder (Syloid® 244) tagged with tetraethylene glycol, uranine, and Tinopal®. A tetraethylene glycol/Tinopal®/uranine liquid aerosol solution is made by mixing 167 g of Tinopal® with 500 g of uranine, and 3.33 L of tetraethylene glycol. To tag the solid aerosol, 300 g of the mixture is slowly added to 300 g of Syloid® 244 amorphous silica while constantly mixing in a Waring-style blender. The product will be a dry powder. The aerodynamic mass median diameter of the solid aerosol must be 2 to 3 µm with a geometric standard deviation of 2 to 3 (a lognormal particle-size distribution) when dispersed in the exposure chamber. The aerosol must be homogeneously dispersed in the wind stream.

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(b) To disperse the fluorescent-tagged powder into the exposure chamber, it is first loaded into a hopper of a screw-feeder-type powder feeder. The output of the dust feed is entrained into the intake airflow of a high-volume blower positioned in the wall of the exposure chamber directly behind the chamber's fan intake. The high turbulence and sheer forces within the high-volume blower disperse the powder into an aerosol. Passage through the fan (plus recirculation through the fan) provides a uniform dispersion of the aerosol throughout the exposure chamber.

(3) Chamber Concentration Monitors

(a) Challenge aerosol mass concentration is to be measured using two filter samplers located in the upstream vicinity of the TP. Sample filters should be at least 98 percent efficient, on a mass basis, for collecting the challenge aerosol.

(b) The filters must be placed in in-line filter holders that are equipped with a sampling probe. The sampling probe diameter must be designed such that the air velocity in the sampling probe is the same (within 20 percent) as the test velocity (isokinetic sampling). The probes must be aligned parallel to the flow field and approximately 0.9 m (3 ft) above the floor, 0.9 m (3 ft) upstream of the TP (when stationary), and approximately 0.3 m (1 ft) left or right of center. Starting and stopping sample collection must be performed remotely.

(c) The mass of aerosol collected on the filters is to be determined by fluorometric analysis.

(d) The duration of the sample collection (typically 30 minutes) is to be measured with a laboratory timer.

(e) The complete measurement technique must be capable of measuring the aerosol mass concentration with an accuracy of ±10 percent, and the samples must be collected for the complete duration of the test.

(4) Temperature, RH, and wind speed inside the chamber must be monitored.

(5) If the protective suit does not include a communication system, or if the communication system within the suit is incompatible with available communication systems, a means must be provided for the TD and the TP to communicate with each other. This may be via hand signals and computer monitor.

3.2.3 Test Preparations

a. Several actions are required before testing begins to ensure reduction of test bias on the part of facilities, instrumentation, or TPs. These actions include:

(1) Testing and certification of:

(a) Test chamber.

(b) Instruments and instrument locations.

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(2) Verification of the sampling and analysis process.

(3) Background testing of TPs, masks, and other test items.

b. Test chamber instrumentation will be calibrated IAW recommended manufacturer's and local procedures, local SOPs, and QA/QC procedures.

c. An ORI will be conducted before the start of the actual test to ensure that all system components are performing adequately.

d. Requirements and criteria for the test chamber and test instrumentation are in Paragraph 2.

e. Test Items

(1) The following steps will be taken during the receipt inspection:

(a) The interior and exterior of each item of clothing must be inspected for rips, tears, and other damage.

(b) Each suit component will be assigned and marked with an individual TIIN. Care must be used to ensure that the TIIN is readily visible and does not compromise the integrity of the test item nor interfere with sampling methods.

(c) Each ensemble will be given a TICN IAW the overall TICN numbering system. The TICN assigned to each item will uniquely identify it by test program and be used to track the item throughout the test process.

(d) Photographic records (with metric scale and in focus) of items showing damage will be made with a written record of the damage. This record will show all areas of damage and will be cross-referenced to the recorded TIIN and TICN. Receipt inspection will be carried out as closely as possible to the time of the test.

(e) If worn suits are to be tested, the wear history of each must be recorded in the chemical test database.

(f) Required maintenance operations described in applicable technical manuals will be performed and all mechanical components tested for operability.

(2) Any pretest environmental conditioning of the clothing/suit required by the SEP and the DTP will be performed.

(3) After the receipt inspection, the suits will be stored in a manner preventing contamination by any chemical vapors present in the storage area.

(4) After testing, each ensemble will be stored individually for future reference (such as additional information on suit size, condition, etc) through the end of the project.

(5) The TD will ensure that the suit developer provides NET whenever necessary.

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f. Dress and Undress Areas

(1) The dress and undress areas will be free of potential interferents.

(2) All TPs will be sized and fitted with the garments to be tested.

3.2.4 Preliminary Actions

a. Pretest actions are designed to ensure that the test is conducted in a test chamber with well defined airflow and controlled temperature and humidity.

b. Test Setup

(1) The aerosol generator and temperature, RH, and wind speed instruments will be placed at the same locations as when the pilot test/ORI was conducted.

(2) The control and signal cables will be neatly arranged to lead from the test chamber to the operator's area.

(3) The instruments for measuring the temperature, RH, and wind speed in the test chamber will be calibrated and installed.

(4) The automatic data collection system software will continuously record the output of the air temperature, air pressure, and RH measuring instruments.

(5) The control system and automatic data collection system software will be connected to all equipment and test instruments and checked for readiness.

(6) The software used for data collection of nonreal-time data will be checked for readiness by performing the entire data collection process using a test (dummy) data set.

4 TEST PROCEDURES

4.1 MIST

4.1.1 Safety

a. The primary emphasis in testing must be on safety. CWA simulants must be handled with care.

b. Tests using simulants will only be conducted IAW the approved SOPs of the testing installation and the procedures specified in the DTP.

c. All procedures must be written in an SOP that has been reviewed and approved by the responsible organizations.

d. All TPs must understand the SOP and test-specific procedures.

e. The TD will ensure that all TPs are thoroughly familiar with the suits to be tested, test procedures, test exercise protocol, and the safety release constraints (if any).

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f. The required MSDS(s), testing protocols, and safety procedures will be on hand at the test site.

g. The HUC has determined that the procedures in this TOP have had sufficient review so that further review is not necessary. However, notification and confirmation of the exemption is still required. Review is required only if deviation from this procedure is considered (Reference 1).

h. Each TP will be informed of potential safety and health hazards involved in test conduct and the precautions required to prevent accidents. A DTC safety release will be provided for each test requiring military personnel as TPs.

i. Each TP must submit to a physical examination, must be certified by a medical authority for eligibility to perform the TP assignment, and must voluntarily sign a TP informed-consent affidavit (Appendix C) before participating in the test.

j. During both the work and rest cycles water/drinking opportunities must be provided throughout the test.

4.1.2 QA/QC

a. A scheme for clearly labeling all test components will be developed.

b. Suit components, samplers, sampling locations, sampler sequences (time on and time off), and raw analytical data must be labeled in a manner precluding misidentification.

c. A second person will independently verify all labeling and markings before the test begins.

d. All test support data will be entered into the proper database before the test begins.

e. Samples

(1) Several of the same kind of samplers will be stored with the test samples as storage controls (Table 2). Some samplers will be blank, and others will be spiked at the concentration range expected to be measured beneath the suit.

(2) Storage control samples will be analyzed with test samples to determine if any sample degradation or interfering chemicals were introduced during storage.

(3) Samples will be stored in a freezer (≤4°C) if they cannot be analyzed within 4 hours of removal. In any case, samples must not be stored longer than 30 days before being analyzed.

f. Chemical Analysis

(1) The chemical analysis procedure will be conducted with an appropriate number of standards, blanks, and analytical controls (10 percent of the samples as a minimum) (Table 2).

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(2) These actions will ensure that the analytical procedure is reliable and will document the precision obtained from analysis of each batch of test samples.

(3) The standards need not be at equal concentration intervals; rather, they should be spaced closer together near the low concentration end of the calibration curve.

(4) Results from the analysis of PAD field QC controls (spiked and blank samples) will be reviewed to determine the amount, if any, of background, interferent, and cross-contamination that existed during testing (Table 2).

g. Each RTM will be checked daily at the testing site with three MeS solutions of different known concentrations (standards) to ensure that the instrument is operating correctly. These data will be plotted on a QC chart showing day-to-day variability. The chart will also indicate whether the instrument is operating within statistical control. At the conclusion of the trial, each RTM will be checked with at least two standards of different concentrations to demonstrate any deviation in the standardization between test beginning and ending levels.

h. QA/QC for Each Trial

(1) Each trial will have at least four QA/QC controls, two spiked and two blanks, per TP. The controls will be deployed in two ways:

(a) The spiked PAD controls will have known amounts of MeS added. The controls will be transported to the test site and returned to the chemical laboratory with the remainder of the samplers for analysis.

(b) The blank PAD controls will be transported to the test site and returned to the chemical laboratory with the samplers for analysis.

(2) Additional samplers may be placed in the dress and undress areas to monitor the level of MeS vapor. These samplers will be returned to the chemical laboratory for analysis.

4.1.3 Pretest

a. The DTP will be approved in advance.

b. An ORI will be conducted and the data reviewed for completeness and accuracy before actual testing begins. The ORI will be conducted in two steps.

(1) Chamber and Test Instrumentation

(a) The chamber and test instrumentation will be tested by bringing the chamber to operational conditions, including temperature, RH, and challenge concentration of MeS. These conditions will be monitored for the length of time required to complete a test trial.

(b) During this period, a set of samplers may be deployed on the outside of an impermeable suit while the wearer enters the chamber and remains for a specified period.

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(c) In the case of an active sampling requirement, an independent referee system may be used to verify that the concentration of MeS vapor in the chamber is within acceptable limits.

(d) Instrumentation in the dress and undress areas will also be checked during the ORI.

(e) Upon completion of the ORI, the data obtained will be collected and transferred to the test data management system database using the same process employed during the actual testing.

(f) The data will be graphed and reviewed by the TD.

(g) Each graph will include: (1) the data collected, (2) variability of the data, and (3) the required control limits.

(2) <u>Suit Sampler Deployment and Analysis</u>

(a) Suit sampler deployment and analysis will be validated by dressing each TP in a clean test ensemble and conducting a 2-hour wear scenario in a clean environment.

(b) This test will be conducted after the TPs have been trained in the use of the test system and in required procedures for personal hygiene.

(c) TPs will undergo the same routine as required for a test, including medical screening, pretest, and showering.

(d) Two participants will be fully instrumented with samplers and will perform a predetermined test regime.

(e) The remaining wearers will be instrumented with five samplers: scalp (21), armpit (26), crotch (34), glove (92), and boot (93) (Figure 1) and will perform the same predetermined test regime.

(f) Sampler removal, dress and undress sampling, and egress procedures will be conducted in the same manner as in testing.

(g) Results from the scalp (21), chin (24), nosecup (90), armpit (26), and groin (33) samplers will be reviewed by the TD to demonstrate that the cleaning and hygiene protocol, as well as the analytical process, are operating correctly. Figure 1 depicts the scalp, chin, nosecup, and groin locations.

(3) The preceding ORI process is adequate preparation to begin testing. The integrated team may, however, direct a pilot test be conducted in place of the pretest if a novel test program, new samplers, or a system with a high potential for interference is to be tested. The pilot test will be conducted only if there are sufficient test items available.

c. <u>Chamber Air Sampling</u>

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(1) As part of the ORI, the air in the chamber will be sampled and analyzed by MIRAN®s for 0.5 hour to determine the presence/absence of background vapor interferents. The test chamber will be operating at the environmental conditions required for the test.

(2) PADs must also be placed in the chamber with the MIRAN®s to determine if there is any interference with GC analysis.

(3) The test can proceed even if there is measurable MeS vapor in the chamber.

(4) If interferent is found in the chamber, the sampler and airstream will be checked to determine if the interferent impacts the analysis.

(5) The MIRAN®s must be corrected for the background interferent if it does not impact the analysis.

(6) If the interferent interferes with the analysis, the chamber must be purged until background sampling indicates that the interferent is less than the LDL for MeS (about 50 ng), depending on the sampling method.

d. EMTs with life support equipment will be present during every test. Before each test, the EMTs will measure and record the physiological signs of the TPs. Physiological signs will include pulse, blood pressure, and body temperature.

e. All TPs must be informed that they can terminate participation in the test at any time.

f. The participants will be dressed in the suits to be tested; the TIIN of the suit worn by each TP will be recorded.

g. Each suit must be the correct size for each TP and must be checked to ensure that it is correctly worn.

4.1.4 Method

a. This TOP is written so that MeS is the CWA simulant and PADs are used as samplers. Although the safety procedures necessary when using a CWA simulant are not as stringent as those required when working with toxic CWAs, the procedures must address all requirements for exposing TPs to MeS.

b. The TPs will be administered a core temperature pill. The pill will take from 30 minutes to 2 hours to become active and stabilize. Activation and stabilization will be monitored by test operators. Once a pill becomes active, the test subject will be moved to the dressing area.

c. In the dress area, test operators will attach a PVSMS and will install the samplers on each test participant at the proposed locations shown in Figure 1.

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d. TPs will be dressed in either the test suit or a standard protective suit to be used for comparison as specified in the DTP. TPs will also wear specified footwear, gloves, protective mask, and hood.

e. Test operators will assist the TPs in obtaining correct closure of the suit at all locations. Each suit will be inspected to ensure proper wear.

f. Generally, each test program will include test and comparison suits.

g. The number of trials specified in the DTP must be adequate to address all experimental conditions with sufficient statistical validity.

h. Temperature, RH, and wind speed instruments, as well as the RTMs, will be placed in the chamber at the locations defined in the mapping phase.

i. All instruments will be started and the test chamber brought to the required test conditions. The output of all instruments will be recorded by the automated data collection software. The chamber atmosphere will be monitored for 0.5 hour for the presence of any interferent(s).

j. The test chamber will be closed and sealed and a negative pressure established. The MeS vapor generator will produce a vapor challenge cloud at the required challenge level.

k. Simulant

(1) The amount of MeS required to generate the vapor concentration specified in the DTP will be dispensed.

(2) Vapor concentration inside the test chamber will be monitored by calculating the average of the readings of the RTMs.

(3) When the vapor concentration decreases by more than 5 percent below the target concentration, additional vapor will be dispensed to maintain the challenge concentration within ±10 percent of the target value.

(4) A feedback control system is recommended for maintaining a constant vapor concentration. The vapor concentration must remain within ±10 percent of the target value during the entire test period for an acceptable trial.

l. TP Monitoring

(1) During every trial, each TP will be continuously monitored by the PVSMS, EMT, on-floor supervisor, and chamber operators for adverse physiological symptoms such as disorientation, hyperventilation, or breathing difficulties.

(2) If such reactions occur, the TP will be removed from the test and given appropriate medical aid.

(3) Core body temperature and pulse rate at which TPs will be removed from the test will be established by the medical staff/EMT before testing begins.

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m. The on-floor supervisor (dressed in protective clothing) will enter the test chamber after the specified MeS vapor challenge concentration has been established and the participants are ready to begin the test.

n. TPs will then enter the test chamber from the antechamber in pairs and at a specified time interval. The time interval will be twice the rotation interval. Generally, the rotation interval is 5 minutes at each station. TPs rotate through each station three times, for a total trial time of 2 hours.

o. TPs will follow the protocol of exercises specified in Appendix B (or the DTP).

p. The activities of the TPs will be visually monitored by the on-floor supervisor, EMTs, test operators, and video cameras.

q. The calibration of the MIRAN®s will be checked three times per day with standard solutions of the simulant.

r. Chamber Control Sampling

(1) Generally, MIRAN®s will be used for chamber control sampling and to obtain an estimate of total challenge. The MIRAN®s are real-time samplers and continuously monitor the challenge concentration in the chamber throughout the duration of the trial.

(2) PADs may be used as a supplemental method to validate chamber control sampling.

(a) This chamber control sampling will be accomplished by placing a TP in an impermeable suit (Tyvek®) and affixing one full set of PADs to the outside of the suit (Table 2).

(b) When all TPs are in the chamber, at the discretion of the test officer (TO), the TP wearing the Tyvek® suit with PADs mounted on the exterior will enter the chamber and will move about for 15 minutes (or other designated time), stopping by each station (Appendix B) and then exiting the chamber.

(c) These PADs will be removed and returned to the chemical laboratory for analysis and the data used to validate the RTM data on the MeS vapor concentration in the chamber for that trial.

s. Posttest Activities

(1) At the conclusion of the specified exercise cycle, pairs of TPs will exit the test chamber and enter the decontamination area where they will decontaminate their boots, gloves, masks, and other decontaminable items with hot soapy water.

(2) TPs will then enter the first-stage undress area that will be monitored for MeS vapor by a low-level RTM and/or a PAD.

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(3) Test operators will remove the protective suits and inspect the samplers for any sign of shifting orientation or position on the TP. These discrepancies will be noted.

(4) The TPs will then enter the second stage undress area. The chamber will be monitored for MeS vapor by a low level RTM and, optimally, a PAD.

(5) The samplers will then be removed and inspected by test operators for any sign of shifting orientation or position on the TP.

(6) The test operators will remove the samplers, check them for correct identification, and place them in carrying containers for transfer to the laboratory for chemical analysis. Any discrepancies must be noted.

(7) QC personnel will randomly add control samplers spiked with known amounts of MeS to the test samplers before transfer to the laboratory.

(8) The TPs may be interviewed at the conclusion of the test and asked to complete a human factors engineering (HFE) questionnaire concerning comfort and fit, ability to communicate, visibility, and ability to perform the exercise protocol.

(9) TPs will be given an opportunity to comment on safety and health hazard issues and to suggest possible improvements to the suit. An HFE questionnaire can be prepared by using TOP 1-2-610 (Reference 2).

4.2 Aerosol

4.2.1 Safety

a. The primary emphasis in testing must be on safety.

b. All procedures must be written in an SOP that has been reviewed and approved by the responsible organizations.

c. All TPs must understand the SOP and test-specific procedures.

d. The TD will ensure that all TPs are thoroughly familiar with the suits to be tested, test procedures, and test exercise protocol.

e. The required MSDS(s), testing protocols, and safety procedures will be on hand at the test site. Approval or exemption and notification from HUC must be obtained before testing begins.

f. Each TP will be informed of potential safety and health hazards involved in test conduct and the precautions required to prevent accidents. A DTC safety release will be provided for each test requiring military personnel as TPs.

g. Each TP must submit to a physical examination, must be certified by a medical authority for eligibility to perform the TP assignment, and must voluntarily sign a TP informed-consent affidavit (Appendix C) before participating in the test.

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h. Water and drinking opportunities must be provided throughout the test.

4.2.2 QA/QC

a. A scheme for clearly labeling all test components will be developed.

b. The ensembles and test samples are to be handled as described in the DTP. The necessary precautions will be taken to ensure that contamination or destruction of the test samples does not occur.

c. Suit components, sampling locations, sampler sequences (time on and time off), and raw analytical data must be labeled in a manner precluding misidentification.

d. A second person will independently verify all labeling and markings before the test begins.

e. All test support data will be entered on appropriate test run data sheets.

f. The donning and doffing procedures to be used must be detailed in the DTP and/or SOP. This includes ensuring that all closures are properly secured and the ensemble is properly worn. Details such as buttoning the top shirt button, wearing the collar up or down, and tucking the shirt sleeves inside or outside of the gloves, etc., are all important. When doffing the ensemble, the main concern is to avoid contamination of the skin with aerosol from other sources (aerosol from the outer garments, contamination from the sampling tube, or contamination from the test operator). Therefore, an assistant should help the TP undress, using scissors to cut off any clothing that would need to be pulled over another body part. Proper handling of the garments and equipment will be necessary to minimize the potential for contamination.

g. Ensembles (new and used) must be sealed from contaminants during storage.

h. If it is necessary to reuse garments, the garments must be laundered IAW Formula II, Department of the Army Field Manual FM 42-414 (Reference 3), or equivalent. Laundering times, temperatures and cycles are listed in Appendix D. Before reuse, the contamination levels in the garments must be verified to be similar to those of the unused garments.

i. The appropriate number of blanks and standards must be used to verify the accurate operation of the various measurement instruments and/or to provide a calibration curve. The calibration methods to be used must be detailed in the DTP.

j. The collection efficiency of the concentration measurement device selected for use must be known.

4.2.3 Pretest

a. The DTP and the conditions of human participation will be approved in advance.

b. The chamber operator will perform one or more trial runs (without a TP) to ensure that the desired aerosol concentration over time and chamber temperature, humidity, and wind speed can be reliably achieved.

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c. An ORI (dry run – no aerosol exposure) will be performed to evaluate the readiness of all systems. The ORI will include background sampling of a TP; donning a mock garment; facemask fit testing; operating the exposure chamber with aerosol, temperature, humidity, and wind speed all at target levels; operating the mass-loader filter samplers; doffing of the mock garment; skin sampling; and fluorometric analysis of all samples. An SOP must be written, approved, and observed before an ORI is performed.

d. The performance of additional dry runs may be conducted as deemed necessary by the principal test operator.

4.2.4 Method

a. The trials will consist of a human subject wearing a correctly-donned, complete chemical-protective clothing ensemble in a test chamber while exposed to a fluorescent-tagged, nontoxic, aerosol simulant. The TP will perform a motion routine (Appendix B) for the entire duration (30 minutes) of the trial. After the trial, the TP's skin will be analyzed for the amount of fluorescent tracer present at various locations. The rate of aerosol deposition to the various sampled areas can then be computed from the amount of fluorescent tracer present. These measurements can then be used to determine a relative protection rating for the ensemble. The use of a fluorescent tracer allows the measurement of minute quantities of aerosol, while minimizing the interference from background aerosol and dust.

b. Clean filters will be inserted into the filter holders, and then slides will be inserted into the stages of the impactors that will be used to measure the concentration and size distribution of the aerosol. Care must be taken to ensure that the slides are properly labeled, carefully handled, and properly positioned. All flow-measurement devices and monitors must be observed to ensure that they are functioning properly. The wind speed, RH, and temperature controllers for the wind room will be set to the required settings and the blower turned on.

c. The TP's skin will be sampled to measure the background fluorescence before each aerosol exposure trial. Samples are to be collected from the body regions that will be sampled following aerosol exposure. The sites for background samples will be determined in the DTP. If, based on experience and the expected levels of aerosol deposition on the skin for the test, the background fluorescence levels are too high, the TP must shower to reduce the background. Samples are to be collected in the identical manner as those collected after an aerosol exposure test, except the sample locations are offset so as not to overlap the previous sample location.

d. While the conditions in the wind room equilibrate, the test garment must be removed from storage and donned by the TP. Clothing items to be worn beneath the test garment will be specified in the DTP. (Typically, the TP wears only a T-shirt, briefs, and socks before donning the ensemble.) The principal test operator will note on the test run data sheet that the complete ensemble is properly donned and will note any taping or anything that does not represent doctrine wear. The test operator will also ensure that the ensemble is properly fit on the TP and that the TP has been informed of the proper execution of the motion routine.

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e. When the TP is prepared and the test chamber has reached the conditions outlined in the DTP, the TP will enter the exposure chamber and be properly positioned in the wind. The airflows of the filter samplers and the impactor will be initiated simultaneously and the time recorded (t = 0 minutes).

f. The TP will begin the motion routine. The test duration will be 30 minutes. The principal test operator will observe the TP during the entire test. Any significant unusual activity during the test that may impact the results (such as a sudden change in a test condition, the TP slipped and fell, etc.) will be noted in the laboratory record book or test form.

g. After the 30-minute exposure, the airflows of the filter samplers will be terminated simultaneously and the time recorded. (The blower and other wind room controls may also be terminated at this time.)

h. The TP and an assistant will proceed to the undress area, where the ensemble will be carefully removed. The doffing procedure detailed in the DTP must be followed carefully. When the ensemble is removed, it must be transported away from the TP to avoid any contamination. After the ensemble has been removed, the undergarments (T-shirt, briefs, and socks) must be removed in the same manner (described in the DTP).

i. When all of the TP's clothing has been removed, the TP's skin will be sampled to recover aerosol that has deposited. This sampling will be performed by pressing a tube against the skin in the area to be sampled and then adding 20 mL of 0.01 N sodium hydroxide (NaOH). The solution will be washed over the skin for approximately 10 seconds and then pipetted into a clean container.

j. The number and location of the areas to be sampled must be indicated in the DTP. Recommended sampling locations are shown in Figure 2 and described in Table 4. For some areas of the skin (such as the earlobe), a series of three cotton swab samples will be taken rather than using the tube method. Each cotton swab sample will be placed in a container with 20 mL solution of 0.01 N NaOH. All samples should be labeled appropriately before they are analyzed.

k. Approximately 5 mL of each of the samples taken from the TP's skin will be analyzed in a fluorometer in order to determine the mass of aerosol that is present in the sample. The results will be recorded and verified to identify and eliminate any errors in reading or recording the data.

l. Aerosol collection substrates will be recovered from the filter holders and impactors, properly labeled, and stored or transported to the chemical laboratory for subsequent analyses. The mass of aerosol collected on the impactor slides and on the filters will be determined by fluorometric analysis.

m. Temperature, RH, and wind-velocity data will be recorded in the laboratory record book or data sheet, and/or stored electronically for subsequent data reduction and review.

n. Any anomalies with filters, holders, impactors, etc., must be recorded.

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o. All of the resulting data (including the names of any computer files) must be recorded in the laboratory record book or test data sheets for subsequent calculations.

p. All samples (and filters) will be clearly labeled and stored in an appropriate storage cabinet or facility until the sample analysis for the trial has been completed; any anomalies may be retested if needed. When data analysis has been completed, the samples may be discarded. Procedures for disposition of overgarments will be described in the DTP.

q. The TP must shower after each trial.

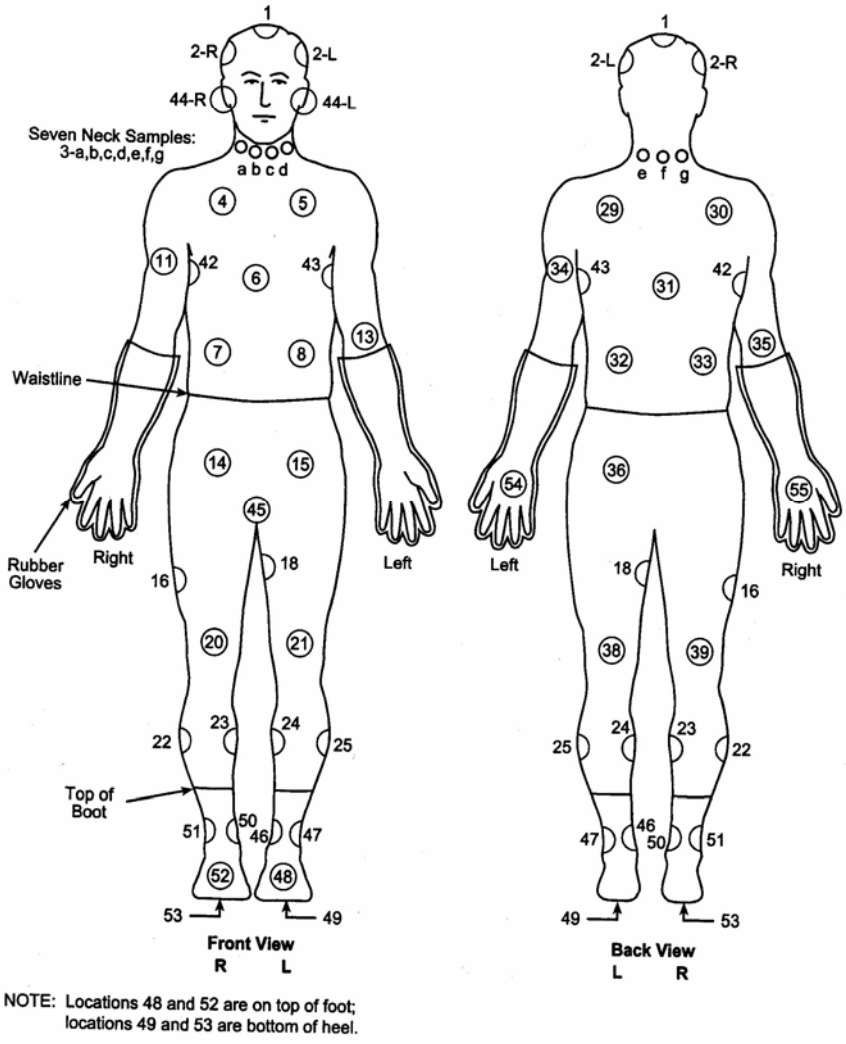


Figure 2. Representative Sample Locations for the System-Level Aerosol Testing; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

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Table 4. Sampling Location Descriptions for System-Level Aerosol Testing; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

Position Number	Description
1	Head, Top
2-R	Head, Right Side
2-L	Head, Left Side
3-a	Neck
3-b	Neck
3-c	Neck
3-d	Neck
3-e	Neck
3-f	Neck
3-g	Neck
4	Upper Chest, Right
5	Upper Chest, Left
6	Mid Chest
7	Lower Chest, Right
8	Lower Chest, Left
11	Upper Arm, Front Right
13	Lower Arm, Front Left
14	Pelvic Area, Right
15	Pelvic Area, Left
16	Outside Upper Leg, Right
18	Inside Upper Leg, Left
20	Mid Leg, Front Right
21	Mid Leg, Front Left
22	Outside Lower Leg, Right
23	Inside Lower Leg, Right
24	Inside Lower Leg, Left
25	Outside Lower Leg, Left
29	Upper Back, Left
30	Upper Back, Right
31	Mid Back

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Table 4. Sampling Location Descriptions for System-Level Aerosol Testing; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits (Cont'd).

Position Number	Description
32	Lower Back, Left
33	Low Back, Right
34	Upper Arm, Left
35	Lower Arm, Right
36	Rump
38	Mid Leg, Back Left
39	Mid Leg, Back Right
42	Side of Torso, Right
43	Side of Torso, Left
44-R	Ear Lobe, Right
44-L	Ear Lobe, Left
45	Scrotum
46	Inside of Foot, Left
47	Outside of Foot, Left
48	Top of Foot, Left
49	Bottom of Foot, Left
50	Inside of Foot, Right
51	Outside of Foot, Right
52	Top of Foot, Right
53	Bottom of Foot, Right
54	Back of Hand, Left
55	Back of Hand, Right

See Figure 2.

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5 DATA REQUIRED

5.1 MIST

The following test data will be provided:

a. Chemical analysis results of all samples, including controls, reported as mass of MeS found in each sample.

b. Concentration values, calculated and reported with flow rates for each sampling system used.

c. Results identified by trial number, TICN, and sampler location on each TP.

d. The individual results and precision obtained from the analytical controls added in the laboratory.

e. Required controls (Table 2).

f. Computer files, graphs, and printouts of the following recorded real-time test chamber data:

(1) Air temperatures.

(2) RH.

(3) MeS vapor concentration data from the MIRAN® s marked with the following:

(a) Date of the test.

(b) Trial duration.

(c) Serial numbers of the instruments.

(d) Locations sampled.

(e) Operator's name.

g. Any notable observations by the test operators (especially system openings, mask breaches, and poor fit).

h. Pretest and posttest physiological data on the TPs.

i. Logbooks from test chamber and laboratory operators and laboratory QA chemist.

j. Completed HFE questionnaires, if required.

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k. Results of the body region analysis procedure (BRAP)* (Reference 4) analysis.

5.2 Aerosol

The following test data will be provided:

a. Chemical analysis results of all samples, including aerosol control samples.

b. Deposition values.

c. Results identified by trial number, TICN, and sampler location on each TP.

d. The individual results for each sample.

e. Any required control results.

f. Computer files, graphs, and printouts of the following recorded real-time test chamber data:

(1) Air temperatures.

(2) RH.

(3) Challenge aerosol-concentration data marked with the following:

(a) Date of the test.

(b) Trial duration.

(c) Serial numbers of the instruments.

(d) Locations sampled.

(e) Operator's name.

g. Any notable observations by the test operators (especially system openings, mask breaches, or poor fit).

h. Pretest and posttest physiological data on the TPs.

i. Logbooks from test chamber and laboratory operators.

j. Completed HFE questionnaires, if required.

k. Results of the BRAP (Reference 4).

*BRAP was developed at U.S. Army Edgewood Research, Development, and Engineering Center (ERDEC), APG. It has been reviewed by the U.S. Army Developmental Test Command Chemical Equipment Permeation Team (APG), the U.S. Army Independent Scientific Panel, as well as toxicology experts at ERDEC.

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6 DATA ANALYSIS

6.1 MIST

6.1.1 Data Review

The data obtained will be reviewed for consistency and acceptability. Specifically, the following will be reviewed:

a. The mass (ng) of simulant detected by the samplers at each sampling position for each TP.

b. Control-data analysis. The chemist must examine the control data (Table 2) to determine if the chemical analysis process was in control. The chemist will approve the data if the process is in control. The chemist will flag the data with a warning that the data are suspect if outside control.

c. The precision obtained on the analytical controls.

d. Printouts and graphs of the following data. These data will be provided to demonstrate that the target values were achieved and maintained within the required tolerance limits during the test.

(1) Recorded results from the MeS challenge concentrations.

(2) Wind speeds.

(3) Air temperatures.

(4) RHs.

e. The interferent control results. These results will reflect any false positive values obtained, as well as how the analytical procedure was altered to discriminate between simulant and interferent.

f. Any variation in TP performance of the exercise protocol as observed by test operators or as shown on the videotape of the actual exercises performed. TPs will be identified by the TIIN of the suit worn.

g. Pretest and posttest physiological data on the TPs.

h. Completed HFE questionnaires, if required.

i. Any notable observations made by the test operators or the TPs.

6.1.2 BRAP

a. The BRAP (Reference 4) will be used to analyze penetration data recorded by the PSDs. The results from the BRAP will be used to compare ensemble performance. The

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absolute assessment of ensemble performance relative to hazards or casualties is not currently possible.

b. Missing data points from the PSDs will be estimated as the geometric mean of the same body point from all ensembles of the same configuration.

c. Penetration data will be entered into the BRAP model and will be normalized to correspond to a challenge concentration of 12,000 mg·min/m^3.

d. The BRAP model then divides the penetration values by the average dose received [the average of the cumulated MIRAN® readings for the three (or more) instruments]. This process produces a performance factor (PF) for each region. These PF data will then undergo two separate analysis regimes:

(1) Localized Effects

(a) To evaluate for localized effects, the PF values will be multiplied by the weighing factor specified in the BRAP for localized effects, corresponding to the effect of distilled mustard (HD).

(b) The resulting local exposure dosage (LED) values will be compared with each other and the lowest value will be selected as the suit performance value. The localized minimum exposure dosage (MED_{HD}) for the configuration is then calculated as the geometric mean of the individual local effects and represents the highest agent exposure dosage to which the ensemble could be externally exposed to HD before the wearer experienced the onset of initial symptoms.

(2) Systemic Effects

(a) To evaluate for systems effects, the PF value for each region will be multiplied by a systemic weighing factor for each endpoint provided by the BRAP.

(b) The resulting values will be summed over all regions and divided into a constant provided by the BRAP in order to establish an overall PF. The systemic agent minimum exposure dosage (MED_{SYS}) is calculated by multiplying this overall PF by another constant provided by the BRAP. The MED_{SYS} for the configuration is then calculated as the geometric mean of the individual systemic effects and represents the highest vapor agent exposure dosage to which the ensemble could be externally exposed to sarin (GB) before the onset of initial symptoms.

e. After the BRAP is complete, an initial analysis of variance (ANOVA) will be performed to check for outliers.

(1) If necessary, a natural log transformation will be performed on the LEDs.

(2) An outlier will be defined as a point having a studentized residual with an absolute value greater than 3.

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(3) If any outliers are identified, those values will be reviewed to determine if there is a discernible cause for each outlier.

(a) If there is a discernible cause, the values will be adjusted, if possible, to reflect the true value; otherwise, the value will be replaced with the geometric mean of the same body point from all ensembles of the same configuration.

(b) If there is no discernible cause for the outlier, the original value will be used.

(4) The outlier analysis will be repeated only once.

f. After outlier analysis is complete, a second ANOVA will be performed.

(1) Penetration data will be analyzed in the second ANOVA to characterize the protective ensembles and to perform comparisons with other ensembles.

(2) The design of the ANOVA will vary by program according to the objectives of the program; however, there will be similarities among the ANOVAs.

(a) The dependent variable will be the minimum effective dosage (MED) or LED of interest calculated by the BRAP.

(b) The main factor(s) will be the variable(s) under question, such as amount of wear or material type.

(c) Factors such as "trial" must be included as blocking factors in order to remove known unwanted sources of variation.

6.1.3 Data Presentation

a. Because of the natural logarithmic transformation performed on the dosage data before analysis, the endpoint values for a particular suit type, mask, hood, etc., will be summarized in terms of geometric means (exponentiation of the least-squares means from the ANOVA) rather than arithmetic means.

b. The number of observations, standard deviations, and 95 percent confidence intervals based on the geometric mean will also be reported.

6.2 Aerosol

6.2.1 Data Review

The data obtained will be reviewed for consistency and acceptability. Specifically, the following will be reviewed:

a. The concentration and particle size distribution of the challenge aerosol to ensure that they were within specifications.

b. The wind velocity, temperature, and RH measurements to ensure that they were within specifications.

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c. The variables relevant to aerosol recovery from the skin, including area sampled, volume of extract, and location identification to ensure proper quantification of mass recovered.

d. Results from the fluorometric analysis and the standards and blanks analyzed with the collected samples to ensure accurate and valid data. If the standards are not within specification to ensure a valid fluorometer calibration, samples are to be reanalyzed.

e. Comments made by the principal test operator during the donning and doffing of the ensemble and during the execution of the test to determine whether anomalous data could be attributed to improper fitting, improper movement, or other test variables.

6.2.2 Data Analysis/Procedure

6.2.2.1 BRAP

a. The BRAP (Reference 4) will be used to analyze penetration data from the fluorometric analysis. The results from the BRAP will be used to compare ensemble performance. The absolute assessment of ensemble performance relative to hazards or casualties is not currently possible.

b. Missing data points from the fluorometric analysis will be estimated as the geometric mean of the same body point from all ensembles of the same configuration.

c. Aerosol deposition velocity (Vd) is calculated by the BRAP model using penetration data.

(1) The rate of aerosol deposition to the TP's skin is expressed as the Vd. The general equation for computing Vd is:

$$Vd = \frac{\text{(Mass of Deposited Aerosol – Background Mass)}}{\text{(Sample Area)} \cdot C_m \cdot T}$$

Where:

C_m = Aerosol mass concentration (typically in mg/m^3)

T = Sample duration (typically in minutes)

(2) Because the fluorescence of the samples is directly related to the mass of the aerosol in the samples, the fluorescence of the samples will be used instead of determining the mass of aerosol in the samples. Therefore, Vd will be computed as:

$$Vd = \frac{\text{(Fluorescence of Deposited Aerosol – Background Fluorescence)}}{\text{(Sample Area)} \cdot C_f \cdot T}$$

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Where:

C_f = Aerosol fluorescence concentration (F/m^3, for example)

(3) For samples where the measured fluorescence is less than or equal to twice the background fluorescence, the Vd will be computed using the background level:

$$Vd = \frac{\text{(Background Fluorescence)}}{\text{(Sample Area)} \cdot C_f \cdot T}$$

(4) This will establish the minimum measurable Vd and preclude reporting zero and negative values (which may occur when the measurements are at the noise level).

6.2.2.2 Local and Systemic Effects Evaluation

The Vds will undergo two separate analysis regimes:

a. Localized Effects

(1) To evaluate for localized effects, the Vd at each end point will be divided into the weighing factor specified in the BRAP (Reference 4) (for localized effects, HD).

(2) The resulting LED values will be compared with each other and the lowest value will be selected as the suit performance value. This minimum value is the MED_{HD}. The MED_{HD} for the configuration is calculated as the geometric mean of the individual MED_{HD} values and represents the highest aerosolized agent exposure dosage to which the ensemble could be externally exposed to HD before the onset of initial symptoms.

b. Systemic Effects

(1) To evaluate for systems effects, the Vd at each end point will be multiplied by the systemic weighing factor provided by the BRAP (Reference 4).

(2) The resulting values will be summed over all regions and divided into a constant provided by the BRAP in order to establish an overall PF. The MED_{SYS} is calculated by multiplying this overall PF by another constant provided by the BRAP. The MED_{SYS} for the configuration is then calculated as the geometric mean of the individual systemic effects and represents the highest aerosolized agent exposure dosage to which the ensemble could be externally exposed to GB before the onset of initial symptoms.

6.2.2.3 Suit Evaluation

a. An initial ANOVA will be performed to check for outliers.

(1) If necessary, a natural log transformation will be performed on the LEDs.

(2) An outlier will be defined as a point having a studentized residual with an absolute value greater than 3.

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(3) If any outliers are identified, those values will be reviewed to determine if there is a discernible cause for each outlier.

(a) If there is a discernible cause, the values will be adjusted, if possible, to reflect the true value; otherwise, the value will be replaced with the geometric mean of the same body point from all ensembles of the same configuration.

(b) If there is not a discernible cause for the outlier, the original value will be used.

(4) The outlier analysis will be repeated only once.

b. After outlier analysis is complete, a second ANOVA will be performed.

(1) Penetration data will be analyzed in the second ANOVA to characterize the protective ensembles and to perform comparisons with other ensembles.

(2) The model used in the ANOVA will vary by program, according to the objectives of the program; however, there will be similarities among the ANOVAs.

(a) The dependent variable will be the MED or LED of interest calculated by the BRAP.

(b) The main factor(s) will be the variable(s) under question, such as amount of wear or material type.

(c) Factors such as "trial day" must be included as blocking factors in order to remove known unwanted sources of variation.

6.2.3 Data Presentation

a. Because of the natural logarithmic transformation performed on the dosage data before analysis, the endpoint values for a particular suit type, mask, hood, etc., will be summarized in terms of geometric means (exponentiation of the least-squares means from the ANOVA) rather than arithmetic means.

b. The number of observations, standard deviations, and 95 percent confidence intervals based on the geometric mean will also be reported.

7 VALIDATION PROCEDURES FOR MIST

7.1 Precision and Accuracy (P&A) Studies

a. The P&A study should be performed annually or after major revisions to the test chamber.

b. Results of the P&A studies will be used to establish performance levels for QC.

c. P&A studies will be conducted with the setup described in the mapping phase section of this document.

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d. The samplers must be standardized and all other instruments must be calibrated before the study begins.

e. The test chamber will be set up with all instruments for measuring temperature, RH, and wind speed.

f. The required number of RTMs will be installed at the locations found to be acceptable in the chamber-mapping phase. The RTMs will be connected to the automated data-collection system, which will collect all test data in real time.

g. The data will be made available for later review and final reporting.

h. Temperature will be set at approximately 21.1°C (70.0°F), RH at 50 percent, and wind speed at 3.2 km/hr (2.0 mph).

i. PADs will be placed in the chamber after the temperature, RH, and wind speed have stabilized. The PADs will sample the chamber for the duration of the test.

j. Background sampling for interferents in the chamber will be conducted for 1 hour, using RTMs and PADs.

k. Interferents found during background sampling will be identified, if possible, and eliminated. At this point, the chamber may be cleaned or the response of the RTM may be corrected for the presence of interferent.

l. Simulant detected during background sampling will not constitute an interferent.

m. After completion of background sampling, simulant vapor will be generated to provide a concentration of 100 mg/m^3. The test chamber and all instruments will be operated for 4 hours, during which more simulant will be added, as needed, to maintain the target vapor concentration of 100 mg/m^3.

n. After 4 hours of measurements, the RTMs will be connected to each other so that the sampling inlets of the second and subsequent RTMs are connected to the exhaust of the previous RTM, with the inlet of the first RTM sampling the air.

o. All RTMs will take measurements on the same sample of chamber air, although at slightly different times.

p. The connected RTMs will be operated for 1 hour.

q. The results obtained by each RTM over the 1-hour period will be compared to determine if the agreement among RTMs is within the goal of ±10 percent.

r. The P&A study will be performed over a total of 4 days. The precision obtained for each of the control parameters (temperature, RH, vapor concentration, and wind speed) will be calculated and the means and limits plotted on QC charts. These calculations will be compared with the performance requirements listed in paragraph 2 of this TOP. The

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results obtained from each RTM at identical times will also be compared to determine if the measured vapor concentration uniformity is within the goal of ±15 percent.

7.2 Chamber Mapping Phase

a. The mapping phase will validate CWA simulant challenge uniformity within the test chamber before testing. A pilot test will be conducted during which simulant will be dispersed. Simulant concentration data will be collected at various points within the chamber during the test to verify that the simulant has been uniformly dispersed. Simulant concentration and, therefore, CWA challenge uniformity will be documented as follows:

(1) A 3-dimensional (3D) grid will be defined inside the test chamber and the chamber will be prepared with all instrumentation and TPs as would be accomplished during an actual test.

(2) A 3D map of the chamber will be prepared using all the instruments, TPs, and/or mannequins that will be used during a real test. The map will include such information as temperature, wind speed, RH, and simulant vapor concentration.

(3) The test will be performed with up to nine TPs placed in the exercise positions outlined in the DTP. The TPs, however, will not be required to perform exercises, and clothed or unclothed mannequins may be used in their place.

(4) After the grid has been defined and the map drawn, a pilot test will be conducted during which the test instrumentation will be connected to the automated data collection system and rotated through all grid positions. The instruments will remain at each position long enough to provide stable readings (a minimum of 10 minutes).

(5) A minimum of three repetitions will be evaluated for the initial mapping. Successive mappings will generally use only one repetition.

(6) The results will be examined to evaluate the variance of each parameter. If the variance for any parameter is greater than the performance requirements listed in paragraph 2 of this TOP, steps will be taken to reduce the variance for that parameter.

(7) When simulant variability throughout the test chamber is within acceptable limits, each instrument will be given a number and assigned a location within the chamber. The same instrument will be used in the same location throughout the testing period.

b. Chamber concentration monitoring devices will be located inside the chamber or will sample the chamber air through the shortest possible lengths of nonabsorbent tubing.

c. TPs/mannequins will be standing stationary at the exercise equipment. The temperature will be set at approximately 21.1°C (70.0°F), the RH at 50 percent, and the wind speed at 3.2 km/hr (2.0 mph).

d. The overall results will be documented in a report. Results will include the measured variances and any decisions made during the test. The positions occupied by each instrument or MIRAN® inlet will be recorded in the test log book.

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e. The exercise equipment specified in Appendix B will be installed in the chamber. When the temperature, RH, and wind speed have stabilized, the RTMs and the real-time data-collection system will be turned on and MeS simulant vapor will be generated to achieve target vapor concentration of 20 to 100 mg/m^3.

f. Data collection will begin when the test operators have departed and the measured parameters have stabilized. Data will be collected for the prescribed length of time (at least 10 minutes), after which the test operators will enter and move all instruments to new designated locations.

g. The location of each instrument will again be recorded. Data will be collected from the new locations for the prescribed length of time. This process will be repeated until the entire test grid has been subjected to all measurements.

h. All data collected will be examined to evaluate the variability of each test parameter throughout the chamber and to establish the initial tolerances for each parameter.

i. Mapping will be required only once. Subsequent trials will require conduct of only one test that can be compared with the original data.

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APPENDIX A. BACKGROUND FOR CHEMICAL VAPOR AND AEROSOL SYSTEM-LEVEL TESTING OF CHEMICAL/BIOLOGICAL PROTECTIVE SUITS

a. To estimate penetration of chemical-protective ensembles by chemical vapor or aerosol, it is necessary to test the entire suit system, including seams, closures, and the areas of transition to other protective equipment (i.e., areas at the ankles, wrists, and neck).

b. The fit or size of the clothing has an effect on penetration. When worn by a person, clothing is subjected to pressure differentials across the garment due to wind, the bellows effect created by movement, breathing, or a combination of the three, all of which may force chemical warfare agent (CWA) vapor or aerosol through the clothing fabric and closures. Bending and moving stresses clothing, particularly over the joints (Reference 5).

c. The inner sorptive layer/reactive layer may partially remove CWA or simulant vapor that penetrates clothing. Perspiration decomposes the CWA or simulant. For aerosol protection, the fabric, if air permeable, must remove aerosols from the air. For permeable and impermeable fabrics, aerosol infiltration at closures must be minimized. The inner sorptive layer/reactive layer is not an effective sink for aerosol particles that have penetrated to the interior of the garment.

d. <u>Human Subjects Versus Mannequins</u>

(1) Static mannequins dressed in protective suits can be tested in CWA-contaminated air but cannot show the effects of physical motion and perspiration. Correlation tests have been conducted using mannequins capable of limited movement with agent. This testing indicates a general agreement in results, but the actual man/system interface can best be tested using actual servicemen and servicewomen wearing the systems. This testing can only be conducted using simulants because the use of agents presents significant safety hazards.

(2) Test participants (TPs) exercising in approved simulant vapor and aerosol environments provide data on the effects arising from physical motion, breathing, and perspiration.

e. Man-in-simulant test (MIST) procedures are described in this test operations procedure (TOP). The MISTs are conducted with either a vapor or aerosol challenge. They are referred to as either vapor or aerosol MISTs.

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APPENDIX B. EXERCISE PROTOCOL

a. For man-in-simulant testing (MIST), exercise equipment will be placed in the test chamber at the numbered positions shown in Table B.1. The test participants (TPs) will move from station-to-station, in numerical order, at the time interval specified in the detailed test plan (DTP) and will perform the exercise required at each station.

b. For aerosol tests, exercises will be performed in the order shown in Table B.2. Each exercise will last for the time interval specified in the DTP.

Table B.1. List of Exercise Stations, Exercises, and Instructions for Man-in-Simulant Test; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

Station	Exercise	Instructions
1	Jumping Jacks	Test participants (TPs) will perform jumping jacks.
2	Sitting Rest	TPs will be seated with right side facing into the wind.
3	Walking Simulation	TPs will walk, facing into the wind, on a treadmill operated at 4.8 km/hr (3 mph) with a 3 percent incline.
4	Sitting Rest	TPs will be seated with left side facing into the wind.
5	Lifting Weights	TPs will move three 10-kg (22.05-lb) weights from one shelf to another shelf (one shelf above the other) and back by (1) bending over, (2) squatting, and (3) kneeling.
6	Taking Cover	TPs will dive under a table, assume a prone position, and stay in that position.
7	Walking Simulation	TPs will walk, with back to the wind, on a treadmill operated at 4.8 km/hr (3 mph) with a 3 percent incline.
8	Climbing Simulation	TPs will climb up a ladder, extend their arms above their heads, and climb down the ladder.

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Table B.2. Exercise Stations, Exercises, and Instructions for Aerosol Test; Chemical Vapor and Aerosol System-Level Testing of Chemical/Biological Protective Suits.

Station	Exercise	Instructions
1	Standing Rest	The first time through the cycle, test participants (TPs) will stand facing the fan. Each time through the cycle, the subject will rotate 90 degrees from the previous position.
2	Walking	TPs will walk back and forth on a 4.88-m (16-ft) long platform, standing tall with arms swinging slightly at their sides.
3	Bending and Reaching	Starting with hands on hips and standing, TPs will slowly bend at the waist, touch their toes, and return to a standing position with hands on hips. Then TPs will extend arms upward above their heads and move heads once up, down, left, and right, and then return hands to hips.
4	Squatting and Reaching	Starting from a standing position, TPs will bend both knees and waist to assume a squatting position. While squatting, TPs will reach up with arms and look up.
5	Trunk Twisting	Starting with hands on hips and standing, TPs will turn their upper bodies to the left 90 degrees while simultaneously extending their arms outward, return to starting position, and repeat procedure while turning 90 degrees to the right.
6	Walking	Repeat Station 2.
7	Running	TPs will run in place. Each time through the cycle, TPs will rotate 90 degrees to the right.
8	Standing Rest	Repeat Station 1.
9	Bending and Reaching	Repeat Station 3.
10	Squatting and Reaching	Repeat Station 4.
11	Trunk Twisting	Repeat Station 5.
12	Walking	Repeat Station 2.
13	Prone Sighting	TPs will lie down on the platform with their heads towards the fan and take a sighting position.
14	Lying on Back	TPs will roll over from the sighting position onto their backs with arms positioned to hold a rifle across the upper chest.

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APPENDIX C. SAMPLE INFORMED CONSENT AFFIDAVITS

1. MAN-IN-SIMULANT TEST (MIST)

INFORM/CONSENT AFFIDAVIT
FOR
MAN-IN-SIMULANT TEST (MIST) PROGRAMS

PROTOCOL OBJECTIVE: Man-in-Simulant-Test (MIST) Programs are designed to test various chemical/biological personal protective ensembles and equipment for possible adoption to provide military personnel with state-of-the-art protection against known chemical and/or biological threats.

INSTITUTION: US Army Dugway Proving Ground

TEST LOCATION: Dugway Proving Ground, UT.

TEST EXECUTION RESPONSIBILITY:
Commander West Desert Test Center
US Army Dugway Proving Ground, UT 84022-5000
DSN/AV: 789-5614
COMMERCIAL: (435) 831-5614

TEST DIRECTOR: A test director will be assigned for each MIST program. This individual has overall management responsibilities for planning, execution, problem solving, data collection, and reporting of the assigned MIST program. Questions or concerns regarding your participation in the MIST program that are not adequately satisfied by the assigned test officer should be directed to this individual for ultimate resolution. This individual will be made known to test participants during orientation briefings.

TEST OFFICER: A test officer will be assigned for each MIST program. This individual is responsible for on-site execution of the MIST program. All activities at the MIST site are under direction of the assigned test officer. This individual will be your day-to-day contact for receiving instructions regarding your personal participation. Questions or concerns regarding your participation in the MIST program should be directed to this individual for resolution. This individual will be made known to test participants during orientation briefings.

1. PROGRAM OBJECTIVES: To obtain sufficient data to characterize the vapor-protective performance of candidate chemical-protective ensembles and equipment. The data from this comparative analysis will support the evaluation of the chemical protection capability provided by the candidate ensembles and equipment in order to further determine acceptance for future military use.

2. GENERAL NOTICE:

a. This research does not expose you, or any test participant, to any chemical or biological agent.

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INFORM/CONSENT AFFIDAVIT
FOR
MAN-IN-SIMULANT TEST (MIST) PROGRAMS

b. All records of this study, including the names and Social Security Numbers of each test participant, will be stored, upon completion of this study, in accordance with AR 340-21.

c. There are no hidden experimental procedures within this study.

d. This research has been reviewed for scientific validity, military significance, and ethical concerns. As a volunteer, you will be authorized, under the provisions of AR 70-25, all necessary medical care for injury or disease which is the proximate result of your participation of this study.

e. The complete plan for this study is contained in the detailed test plan for this MIST program. A copy of the detailed test plan will be made available to you upon request.

f. You have been asked to participate in a research study conducted by the U.S. Army Dugway Proving Ground. It is important that you read and understand the following principles that apply to all participants in our studies:

(1) Your participation is entirely voluntary.

(2) You may withdraw from participating in this study at any time. Refusal to participate will involve no penalty or loss of benefits to which you are otherwise entitled.

(3) After you read the explanation, please feel free to ask any questions that will allow you to clearly understand the nature of the study.

(4) Inclusion Criteria. Subjects in this test must be either active military, government civilian, or government contractor.

(5) Exclusion Criteria. Subjects with the following conditions will be excluded from test participation:

(a) Anyone with a known heart problem.

(b) Anyone with high blood pressure (systolic above 160 or a diastolic above 100).

3. DESCRIPTION OF STUDY: Detailed information concerning the specific nature of the study in which you have been asked to participate is provided below:

a. Title of Test: MIST of candidate individual chemical-protective equipment for future military application.

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FOR
MAN-IN-SIMULANT TEST (MIST) PROGRAMS

b. Purpose: To obtain sufficient data to characterize the vapor-protective performance of candidate chemical-protective ensembles and equipment. The data from this comparative analysis will support the evaluation of the chemical-protection capability provided by the candidate ensembles and equipment in order to further determine acceptance for future military use.

c. Duration: The duration of this program is dependent on the number of protective ensembles to be tested. The current schedule is to test 16 ensembles per trial day with eight test participants in each trial. You will be informed of the number of trial days during the in-brief.

d. Chemical Agent Simulant: The simulant to be used in this program is methyl salicylate (MeS). This chemical is also known as oil of wintergreen which is used in many products that are used both internally and externally. A copy of the material safety data sheet (MSDS) will be provided to you on request.

e. Test Procedure: Each test participant will be wearing prototype or standard chemical-protective garments during a 2-hour mission scenario. Only one wearing or MIST trial per test participant will be conducted in a given day.

4. REASONABLE FORESEEABLE RISKS OR DISCOMFORTS: The risks or discomforts that may arise from this study include heat stress caused by moderate to heavy work performed while wearing the standard and prototype garments.

a. Heat stress may occur during periods of moderate work while wearing the standard garments in hot weather. Water will be made available throughout the test, and participants should drink ample amounts to protect against heat stress. Symptoms of heat stress include: heat cramps of the muscles of the stomach, arms, or legs and heat exhaustion which may cause you to be dizzy, faint, or weak. A test participant experiencing heat stress should notify a buddy, seek shade, drink water, remove outer clothing, lie down, and elevate feet. Although heat stress is not in itself life threatening, if ignored it could lead to a much more serious situation called heat stroke. Heat stress monitoring and egress procedures are incorporated into the MIST to insure the maximum protection for each individual.

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FOR
MAN-IN-SIMULANT TEST (MIST) PROGRAMS**

b. Heat stroke is always life threatening. In heat stroke, the person's temperature-control system that causes sweating stops functioning properly. The body temperature rises so high that brain damage and death will result if the person is not cooled quickly. The main signs of heat stroke are: red or flushed skin; hot, dry skin (although the person may have been sweating earlier); and extremely high body temperature (up to 106° F). There may be dizziness, nausea, headache, rapid pulse, and unconsciousness. If you feel these symptoms or notice a buddy who appears to have these symptoms, immediately notify the on-site floor boss, emergency medical technician (EMT), test officer, test data collector, or a buddy. The test participant will be given immediate first aid and evacuated to the Aid Station or Health Clinic.

c. EMTs will be on-site throughout the test. Test participants will have their core body temperature monitored using the Personal Vital Signs Monitoring System (PVSMS). The purpose of the PVSMS is to help mitigate the occurrence of heat stress during the trials. You will be requested to swallow a disposable monitor pill and wear a small transmitter on your upper arm. The pill has been used in hospitals and in previous field studies and is considered safe. The pill will measure your core body temperature and transmit the measurement to the transmitter using a low-power transmitter. You will also be asked to wear heart rate and activity monitors throughout the test, which will also provide safety-monitoring data. The Army Surgeon General's Office and the U.S. Army Developmental Test Command's Human Use Review Board approved the PVSMS for use during this test. The pill is eliminated from your body through normal waste elimination processes. The pill is processed like any other medicine or food except that the pill does not break down or get digested; it goes in and comes out whole. The pill is a little larger than the average medicine capsule and will only be used once then discarded. More information is available about the PVSMS upon request.

5. PRECAUTIONS TO BE OBSERVED BEFORE AND AFTER THE TEST: Do not drink alcohol for 12 hours prior to participating in this test. This may cause dehydration and lead to heat stress. Drink plenty of liquids (preferably water) before, during, and after participation. Water or Gatorade® will be provided throughout the test trials. Notify the on-site EMT or test officer of any change in your health status. A list of foods and products, e.g., wintergreen-flavored gum and lifesavers, Ben-Gay®, etc., that may not be used prior to testing will be provided. Utilization of these foods/products will have a detrimental effect on the test data results.

6. BENEFITS TO SUBJECTS OR TO OTHERS: The subjects who participate in this test will gain a unique training experience in the operation, wear, and maintenance of the newest chemical-protective garments available and other candidate commercial garments. The results (both comments and statistical data) will be used as input for both upgrading the design and operation of the chemical-protective suits in the future and providing guidance and procedures for safe operation today.

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INFORM/CONSENT AFFIDAVIT
FOR
MAN-IN-SIMULANT TEST (MIST) PROGRAMS

7. ASSURANCE OF CONFIDENTIALITY OF SUBJECT'S IDENTITY: I understand that this study is affected by the Privacy Act of 1974 (top of the Inform/Consent Affidavit) and have been made aware of the safeguards available to me because of that act. I also understand that the information gained from this study may be analyzed and used as part of a scientific publication, but I will in no way be personally identified. I understand that it may be necessary for representatives of the Surgeon General's Human Subjects Research Review Board to inspect my records concerned with this study in their capacity as reviewing officials.

8. CIRCUMSTANCES UNDER WHICH YOUR PARTICIPATION MAY BE TERMINATED WITHOUT YOUR CONSENT: At the discretion of a physician, you may be withdrawn at any time from this study for medical reasons. You will be removed if your core body temperature reaches 38.8°C (102°F). You may also be removed, at the discretion of the test officer, if it becomes obvious that continuation in the study is likely to result in injury to you or others in the study.

9. COMMENTS FROM SUBJECTS: Your safety is our primary concern. Any feedback in the form of questions, comments, and criticism is essential to the success of the test. You will be asked to fill out human factors engineering questionnaires and to comment on safety and health hazard issues.

VOLUNTEER'S INITIALS _____

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INFORM/CONSENT AFFIDAVIT FOR MAN-IN-SIMULANT TEST (MIST) PROGRAMS

IF THERE IS ANY PORTION OF THIS EXPLANATION THAT YOU DO NOT UNDERSTAND, ASK THE TEST DIRECTOR AND/OR TEST OFFICER BEFORE SIGNING.

I concur that the use of the PVSMS has been explained to me and is considered an investigational device. I understand that my signature below is an agreement to use PVSMS and associated pill as a safety monitor.

I have received a copy of the volunteer explanation and volunteer inform/consent affidavit that I may keep.

_____
VOLUNTEER'S SIGNATURE DATE

_____
VOLUNTEER'S PRINTED NAME

_____
PERMANENT ADDRESS

I was present during the explanation referred to above, as well as during the volunteer's opportunity to ask questions. I hereby witness the volunteer's signature.

_____
WITNESS' SIGNATURE DATE

_____
WITNESS' PRINTED NAME

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1. AEROSOL TESTING

INFORM/CONSENT AFFIDAVIT AND EXPLANATION FOR PARTICIPATION IN AEROSOL TESTING

The affidavit and explanation are presented in two parts as follows:

Part 1. "Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs"

This inform/consent affidavit and explanation will allow you to examine the wind tunnel facility, to wear protective garments and mask, and to be in the wind tunnel with the fan running (but with no aerosol). The purpose of this is to provide you with information you may need in order to make an informed decision about participating in subsequent aerosol challenge tests ("Aerosol Challenge of Chemical-Protective Garments").

Part 2. "Aerosol Challenge of Chemical-Protective Garments"

This volunteer agreement affidavit and explanation will allow your participation in the aerosol challenge test program.

Part 1

Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs
General Informed Consent

The "Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs" is a program to allow you to participate in a visit to the wind tunnel and protective garments in order for you to make an informed decision relative to your participation in the aerosol challenge study.

This program is being conducted by the TESTING ORGANIZATION under contract or subcontract with U.S. Government Agencies.

Your participation in "Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs" is completely voluntary and you are free to withdraw at any time. Refusal to participate will involve no penalty or loss of benefits to which you are otherwise entitled.

You are being asked to participate in a visit to the wind tunnel. You will have the opportunity to examine the wind tunnel, wear the protective garments and respirator, and ask any questions you may have. The protective garments include an overgarment that you will wear over

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your clothes, full-face respirator, gloves, and boots. New and previously-worn laundered garments may be tested. No aerosols or chemicals will be used, and your time in the wind tunnel will be limited to not longer than roughly XX minutes. Your participation in this visit will be about X hour(s).

This research does not directly or indirectly expose you, or any study personnel, to any chemical, biological, or nuclear warfare agent. Some people find that wearing a protective respirator gives them a claustrophobic feeling and are, therefore, uncomfortable. Also, wearing the protective garments, hearing protectors, or eye goggles may be uncomfortable for some. Except for these discomforts, there are no known health risks or discomforts associated with this study. The adequacy of safety measures has been examined by the MONITORING AGENCY. Authority to use human volunteers has been granted by the project sponsor.

You will receive no benefit for your participation except gaining insight into what the full-scale tests will involve.

If you have any questions regarding this program, you may contact the Principal Investigator, NAME, at (XXX)-XXX-XXXX. If you have any questions regarding the rights of research subjects or problems resulting from the research, you may contact NAME at (XXX)-XXX-XXXX.

Your signature below indicates that the purposes and procedures of the "Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs" have been explained to you and that you consent to participate. It does not, however, obligate you to participate in any part of the "Examination of the Wind Tunnel Facility and Protective Garments Before Participation in the Aerosol Challenge Test Programs."

_____	_____
Participant's Name - Printed	**Participant's Signature and Date**
_____	_____
Signature of Person Obtaining Consent	**Date**

Part 2

Aerosol Challenge of Chemical-Protective Garments
General Informed Consent

The "Aerosol Challenge of Chemical-Protective Garments" is a study to determine the rate at which wind-driven aerosol particles deposit on the skin and clothing of a person wearing chemical-protective clothing. A wide range of protective clothing configurations will be evaluated. Possible garments include the standard battledress overgarment (BDO), Joint Services Lightweight Integrated Suit Technology (JSLIST) garments, and the chemical-protective undergarment (CPU). Also, the effects of movement, such as walking, reaching, bending, etc., will be evaluated for their effect on aerosol deposition to the subject's skin. During the past years, we have conducted similar tests with protective clothing without

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problems. Information gained from this protocol may be used as part of a scientific publication in professional journals, but you will in no way be personally identified.

This program is being conducted by the TESTING ORGANIZATION under contract or subcontract with U.S. Government Agencies.

Your participation in "Aerosol Challenge of Chemical-Protective Garments" is completely voluntary and you are free to withdraw at any time. Refusal to participate will involve no penalty or loss of benefits to which you are otherwise entitled.

Your participation in the study will involve wearing chemical-protective garments that include a face mask, hood, jacket, pants, and gloves. In some tests, you will wear the same clothing items that you wore previously in wear trials. For other tests, new clothing items will be used. You will then enter a wind tunnel that has been filled with the test aerosol. The test aerosol contains a fluorescent tracer that is used to detect its presence on your skin and in the clothing. Because of the aerosol, the wind tunnel air will be hazy. After you enter the wind tunnel, the fan will be turned on to produce a wind speed of 10 mph. This portion of each test will last 30 minutes. During the test, you will perform a series of motions, including walking, squatting, standing, reaching, twisting, and briefly running in place. At the end of the test, the fan will be turned off and you will leave the wind tunnel. An assistant will help you undress down to your undergarments In a private room, you will remove the briefs and put on a pair of gym shorts. The assistant will then rinse approximately 40 areas of your skin with a dilute rinsing solution.

The aerosol on your skin will, in nearly all cases, not be visible to the naked eye. However, because the aerosol is fluorescent, it may be visible under black light. Your skin will be examined with a black light to help determine where aerosol deposition has occurred. Photographs and videotape may be taken of your body while you are wearing gym shorts in order to document patterns of aerosol deposition. Photographs and videotape may also be taken of you wearing the protective garments to document the garment configuration worn in the test.

Your participation in each test will involve approximately X hours of your time. Your participation will occur over an approximate X-week period currently scheduled for MONTH DAY – MONTH DAY, YEAR. During this period, approximately XX tests will be performed, divided roughly equally between XX volunteers; thus you will perform approximately X of the X tests. You will participate as a test volunteer in no more than one test per day.

This research does not directly or indirectly expose you or any study personnel to any chemical, biological, or nuclear warfare agent. Your exposure will be limited to only those substances that have been reviewed by the project's Medical Monitor. These substances include Syloid® tagged with uranine, tetraethylene glycol, and Tinopal® CBS-X, and 0.01N sodium hydroxide (NaOH). Inhalation of the test aerosol poses a minimal hazard to you; minor skin irritation and/or allergic reaction is possible for a few sensitive individuals. There are no other known health risks or discomforts associated with this study. The adequacy of safety measures has been examined by the MONITORING AGENCY.

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There will be no compensation for your participation in this program. Your participation will, however, help to improve chemical-protective clothing used by the U.S. Armed Forces. Circumstances under which your participation may be terminated without your consent include: health conditions under which your participation would possibly be dangerous, other conditions that might occur to make your participation detrimental to your health, and, at the discretion of the test director (TD), you may be removed if, in his/her opinion, you have demonstrated disciplinary or motivational problems that interfere with your full participation in the test.

If you have any questions regarding this program, you may contact the Principal Investigator, NAME, at (XXX)-XXX-XXXX. If you have any questions regarding the rights of research subjects or problems resulting from the research, you may contact NAME at (XXX)-XXX-XXXX. Questions related to health may be directed to NAME at (XXX)-XXX-XXXX. A copy of the complete plan of this study, called the Protocol, will be available to you upon request.

You have been provided with a copy of a Privacy Act Statement, which describes the safeguards available to you because of the Privacy Act of 1974. You have been given the opportunity to review the Privacy Act Statement, ask questions, and retain a personal copy. You understand that information gained from this protocol may be used as part of a scientific publication in professional journals, but you will in no way be personally identified.

Your signature below indicates that the purposes and procedures of the "Aerosol Challenge of Chemical-Protective Garments" have been explained to you and that you consent to participate.

_____	_____
Participant's Name - Printed	**Participant's Signature and Date**
_____	_____
Signature of Person Obtaining Consent	**Date**

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APPENDIX D. LAUNDERING CYCLES OF FORMULA II FROM FM 42-414

Step	Laundering Cycles	Water Level	Time (Minutes)	Temperature	Supplies
1	Fill	High		90F-110F	
2	Wash	High	5	90F-110F	6 oz. PD245
3	Drain				
4	Fill	High		90F-110F	
5	Wash	High	5	90F-110F	4 oz. PD245
6	Drain				
7	Fill	High		90F-110F	
8	Rinse	High	2	90F-110F	
9	Drain				
10	Fill	High		90F-110F	
11	Rinse	High	2	90F-110F	
12	Drain				
13	Fill	High		90F-110F	
14	Rinse	High	2	90F-110F	2 oz. Sour
15	Drain				
16	Extract		3-5		

Amount of supplies based on a 60 pound load

Tumble Drying: Tumble dry at low temperature setting (110F/43C) and remove immediately upon completion of cycle. Avoid overdrying

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APPENDIX E. GLOSSARY

3D – three dimensional
AATCC – American Association of Textilists, Chemists, and Colorists
ANOVA – analysis of variance
APG – Aberdeen Proving Ground
BDO – battledress overgarment
BRAP – body region analysis process
CB – chemical/biological
CPU – chemical-protective undergarment
CWA – chemical warfare agent
DEM – diethyl malonate
DMMP – dimethyl methylphosphonate
DTC – US Army Developmental Test Command
DTP – detailed test plan
EA – environmental assessment
EIALC – environmental impact assessment for life cycle
EIS – environmental impact statement
EMT – emergency medical technician
ERDEC – U.S. Army Edgewood Research, Development, and Engineering Center
FEP – fluorinated ethylene propylene
GB – sarin
GC – gas chromatograph
HD – distilled mustard
HFE – human factors engineering
HUC – Human Use Committee
IAW – in accordance with
IR – infrared
JSLIST – Joint Services Lightweight Integrated Suit Technology
LDL – lower detection limit
LED – local exposure dosage
MED – minimum effective dosage
MED_{HD} – minimum effective dosage of HD
MED_{SYS} – overall systemic capability of the protective garment
MeS – methyl salicylate
MINICAMS® – Miniature Automatic Continuous Air Monitoring System®
MIRAN® – Miniature Infrared Analyzer®
MIST – man-in-simulant test(ing)
MSDS – material safety data sheet
NaOH – Sodium hydroxide
NEPA – National Environmental Policy Act
NET – new equipment training
NRDEC – U.S. Army Natick Soldier Research, Development, and Engineering Center
ORI – operational readiness inspection

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P&A – precision and accuracy
PAD – passive absorbent device
PF – protection factor
PFA – perfluoroalkoxy
PSD – passive sampling device
PTFE – polytetrafluoroethylene
PVSMS – personal vital signs monitoring system
QA – quality assurance
QC – quality control
REC – record of environmental consideration
RH – relative humidity
RSD – relative standard deviation
RTM – real-time monitor
SAR – safety assessment report
SEP – system evaluation plan
SF_6 – sulfur hexafluoride
SOP – standing operating procedure
SSP – system support package
SSPL – system support package list
SST – solid sorbent tube
TD – test director
TEMP – test and evaluation master plan
TICN – test item control number
TIIN – test item identification number
TO – test officer
TOP – test operations procedure
TP – test participant
Vd – deposition velocity

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APPENDIX F. REFERENCES

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Forward comments, recommended changes, or any pertinent data that may be of use in improving this publication to Test Operations and Workload Team (CSTE-DTC-TM-B) US Army Developmental Test Command, 314 Longs Corner Road, Aberdeen Proving Ground, Maryland 21005-5055. Phone: (410)278-1486, DSN: 298-1486. Technical information may be obtained from the preparing activity: US Army Dugway Proving Ground (CSTE-DTC-DP-WD-C), Dugway, Utah 84022-5000. Phone: (801) 831-5137, DSN: 789-5137. Additional copies are available from the Defense Technical Information Center, John J. Kingman Road, Suite 0944, Fort Belvoir, Virginia 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

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